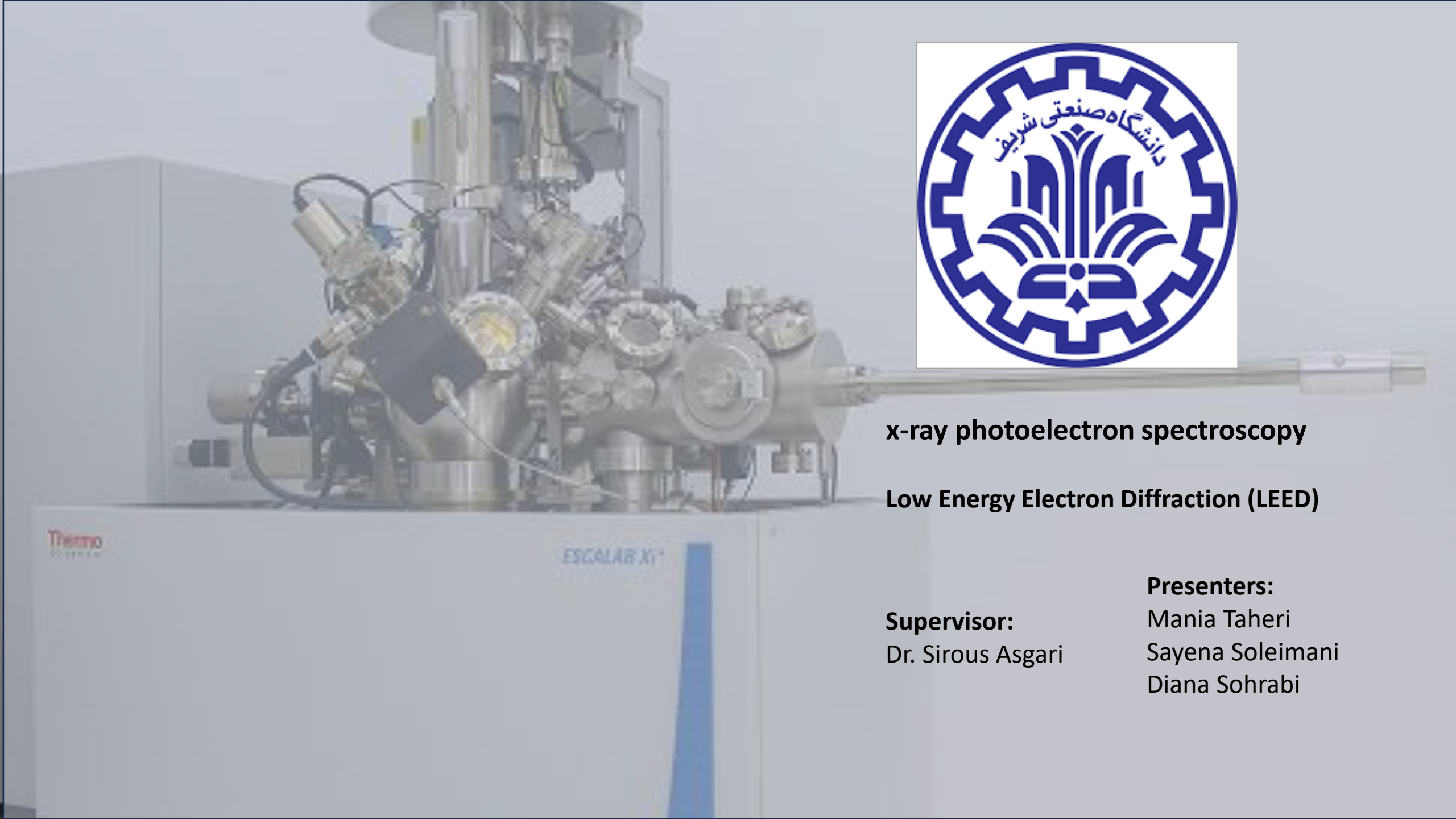


x-ray photoelectron spectroscopy

Low Energy Electron Diffraction (LEED)

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1.Introduction

2.History

3.Fundamental Physics

4.Instruments

5.Understanding XPS Spectra

- Elemental Specificity
- Chemical Sensitivity
- Depth Profiling
- Quantitative Analysis

6.Sample Preparation

7.Advantages and limitation

8.application

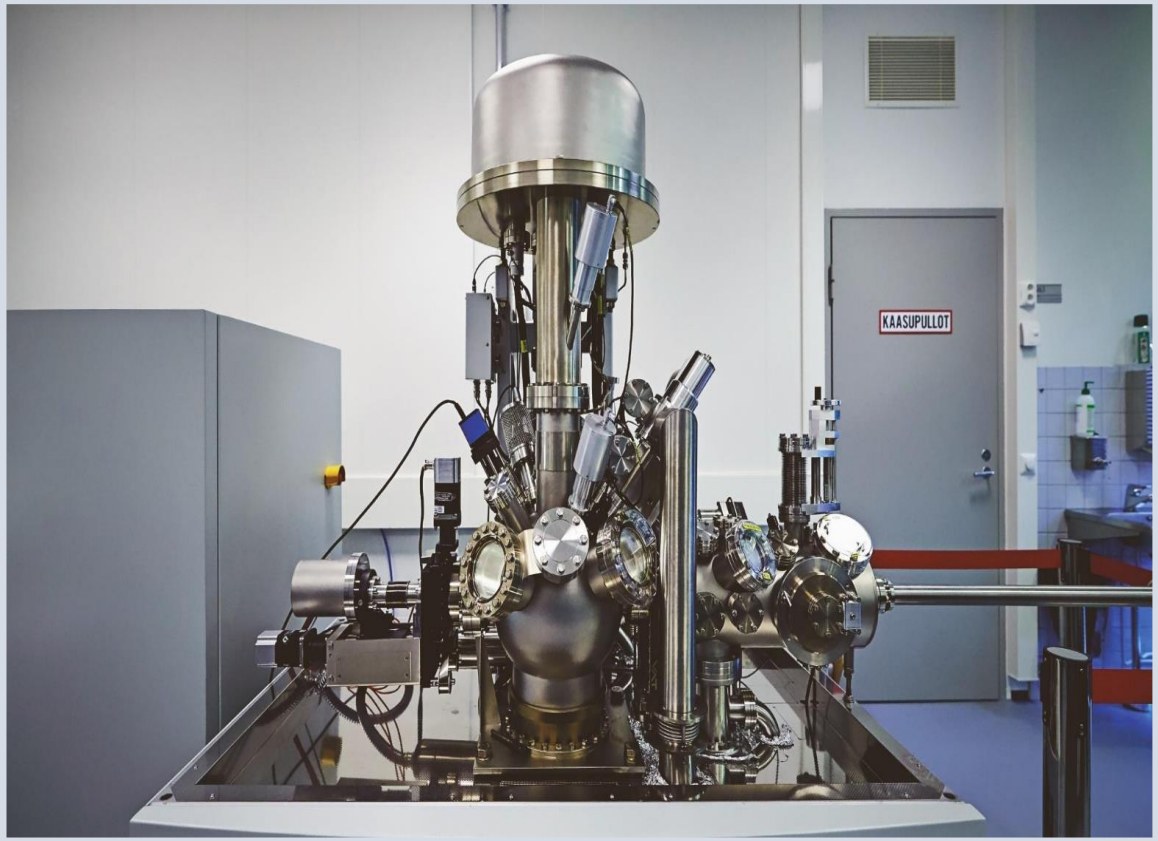
8.Available XPS

Introduction to X-ray Photoelectron Spectroscopy (XPS)

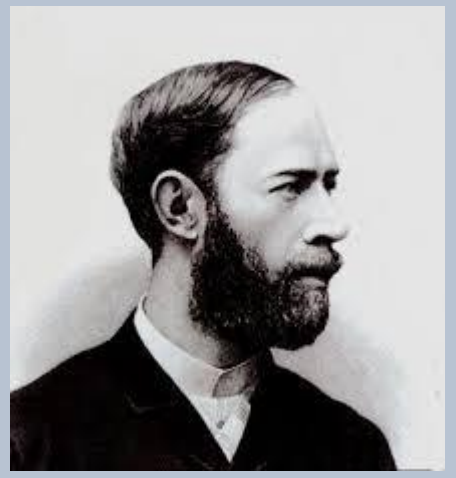
X-ray Photoelectron Spectroscopy (XPS), also known as Electron Spectroscopy for Chemical Analysis (ESCA), is a powerful **surface-analysis technique** that reveals the chemical composition and chemical state of the outermost atomic layers of materials.

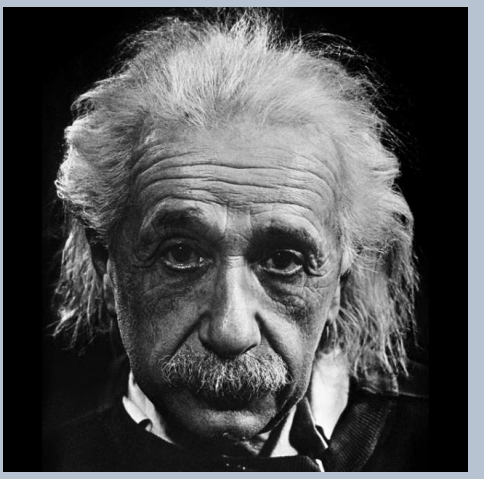
When a material surface is irradiated with monochromatic X-rays, photoelectrons are emitted via the photoelectric effect. By measuring the kinetic energy of these emitted electrons, one can compute their binding energy, which is characteristic for a given element *and* sensitive to its chemical environment (e.g., oxidation state, type of bonding).

Because only electrons from the top few nanometers (typically ~ 3–10 nm) of the surface can escape without losing too much energy, XPS provides highly surface-sensitive information.

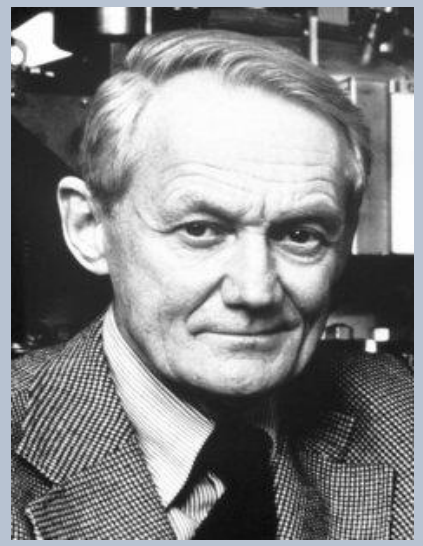


Year / Period	Milestone
1887	First observation of the photoelectric effect by Heinrich Rudolf Hertz — electrons emitted under UV light.
1905	Photoelectric effect explained theoretically by Albert Einstein, introducing the concept of light quanta (photons).
1907	P.D. Innes records broad electron emission bands from a metal under X-ray irradiation — first rudimentary XPS-like spectrum.
1907–1920s	Various researchers (Moseley, Robinson, Rawlinson) attempt X-ray electron emission experiments, but limitations in technology prevent clear, interpretable results.
1954	Kai Siegbahn's group builds the first high-resolution XPS spectrometer and obtains clear spectrum (e.g. NaCl surface), showing the method's potential.
1969	First commercial monochromatic XPS instruments become available — spreading access beyond pioneering labs.
1981	Nobel Prize awarded to Siegbahn for development of high-resolution electron spectroscopy (XPS / ESCA).
1990s–present	XPS usage grows exponentially and becomes the standard technique for surface chemical analysis across material science, nanotechnology, catalysis, thin films, coatings, semiconductors, etc







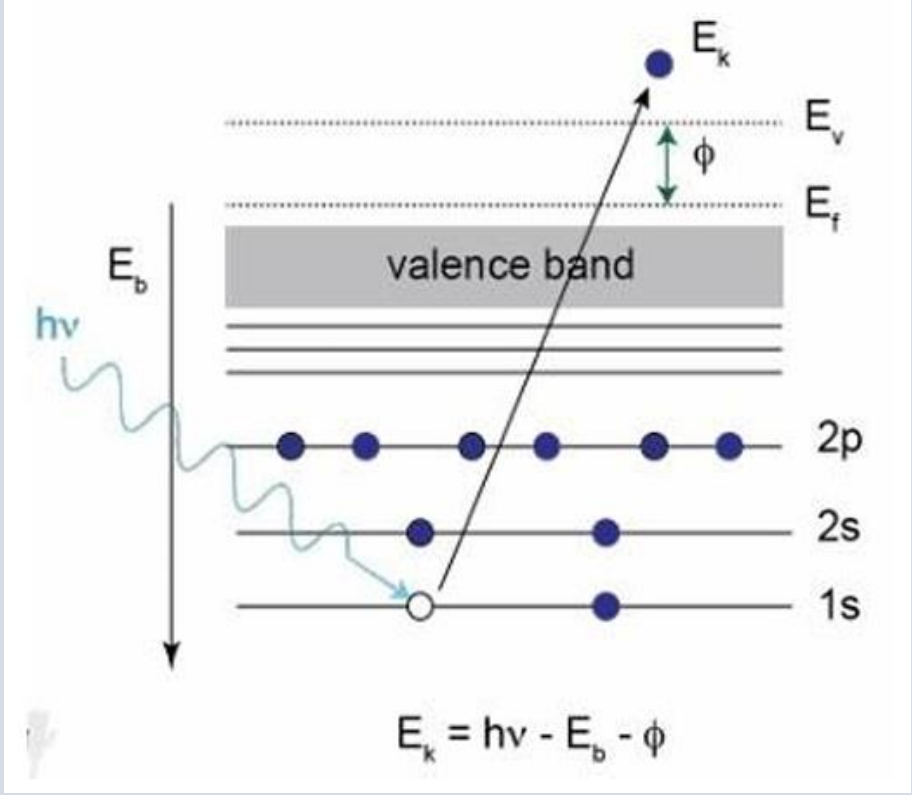


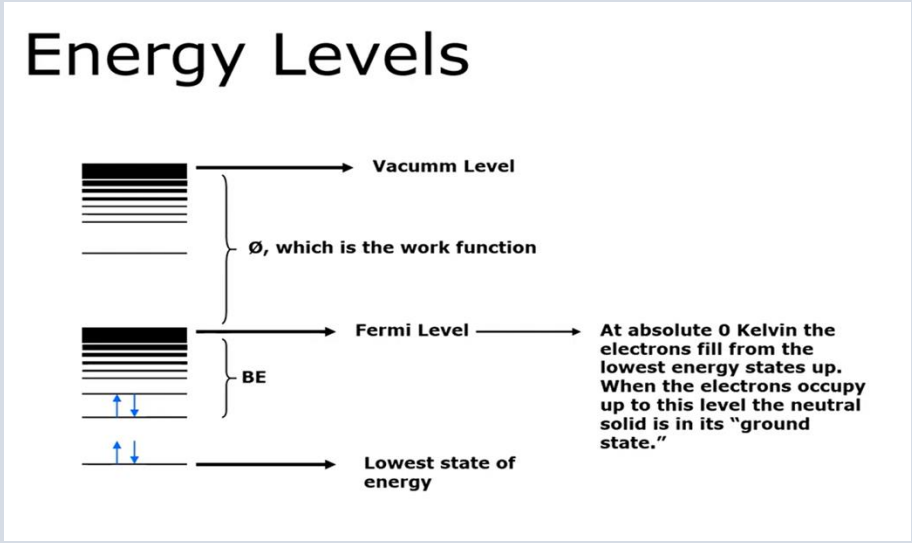
Fundamental Physics Behind XPS

X-ray Photoelectron Spectroscopy (XPS) is based on the **photoelectric effect**, in which absorption of an X-ray photon transfers its energy to a core electron, ejecting it from the atom. The kinetic energy (E_k) of the emitted electron is measured, and from this the **binding energy** (E_b)—the energy required to remove the electron from its atomic orbital—can be calculated using

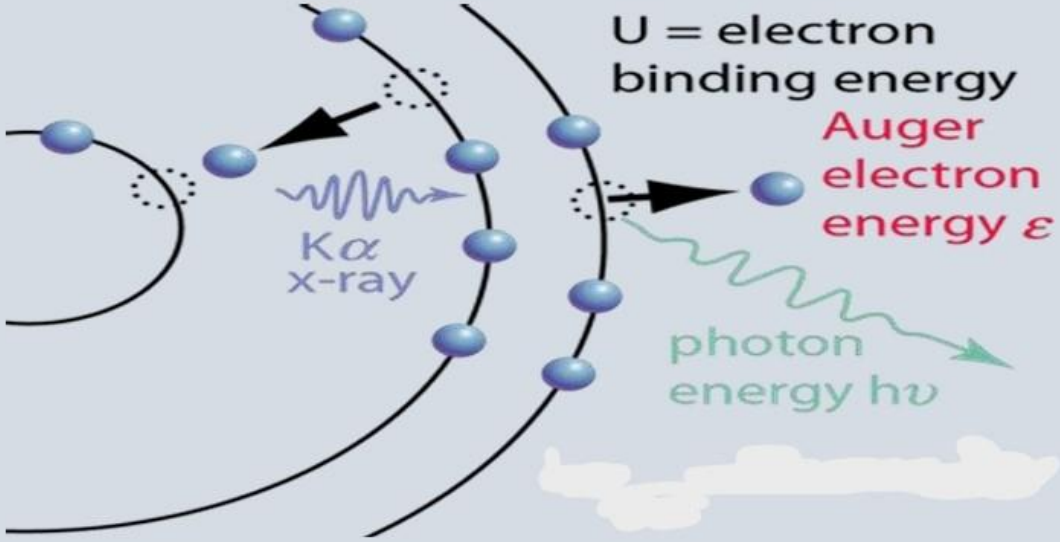
$$E_b = h\nu - E_k - \phi$$

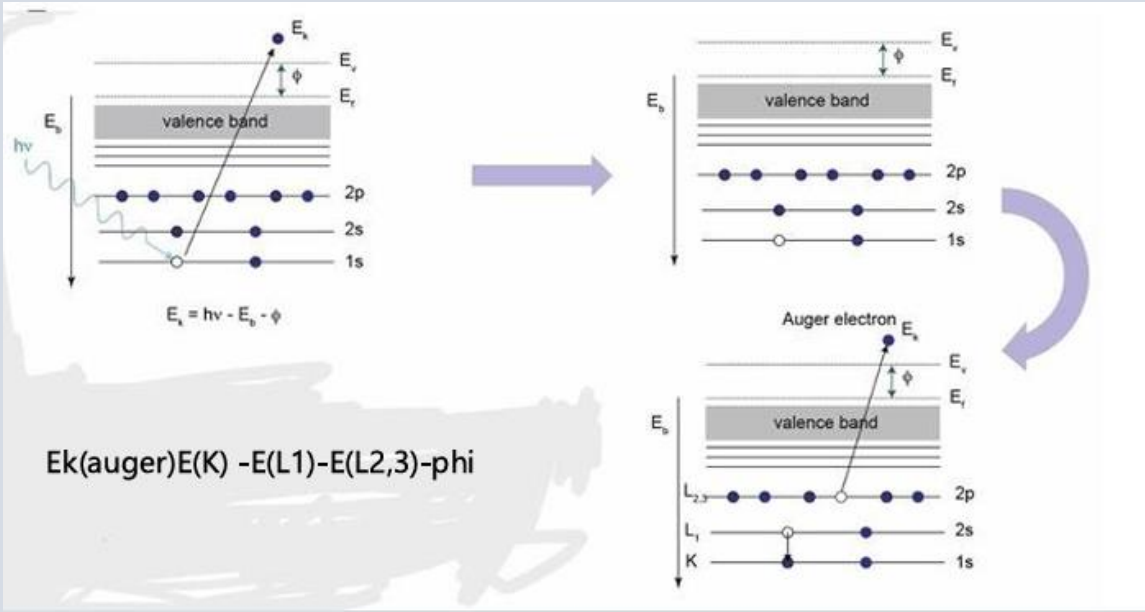
where $h\nu$is the photon energy and $\phi$is the **work function**, the energy needed for an electron at the Fermi level to escape the surface into vacuum. Because **core electrons** are tightly bound and localized near the nucleus, their binding energies are highly characteristic of each element (and sensitive to chemical environment), giving XPS its elemental and chemical specificity. Valence electrons, in contrast, are delocalized and do not provide atom-specific information, which is why XPS focuses on core levels.





A second physical process relevant to XPS is the **Auger effect**. After a core electron is removed, the atom relaxes by filling the core hole with an electron from a higher level, releasing excess energy. Instead of emitting an X-ray photon, this energy can eject another electron, called an **Auger electron**. The kinetic energy of Auger electrons depends only on atomic energy levels and is **independent of the incident X-ray energy**, making Auger peaks useful for chemical analysis alongside photoelectron peaks. Together, the photoelectric effect and the Auger process form the physical basis of XPS and provide quantitative information about elemental composition, oxidation states, and the electronic structure of materials.





Instruments
1.X-ray Source:

1. Twin Anode (Dual-Anode) X-Ray Source

A twin-anode X-ray source is the traditional and most widely used non-monochromatic X-ray generator in XPS instruments. It contains **two different metal anodes**, typically:

Aluminum (Al Kα, 1486.6 eV) Magnesium (Mg Kα, 1253.6 eV)

2. Monochromatic X-Ray Source

A monochromatic X-ray source produces **high-purity, energy-narrowed X-rays** by using an anode (usually aluminum) combined with a **crystal monochromator**, commonly a quartz or silicon crystal cut to satisfy Bragg's law.

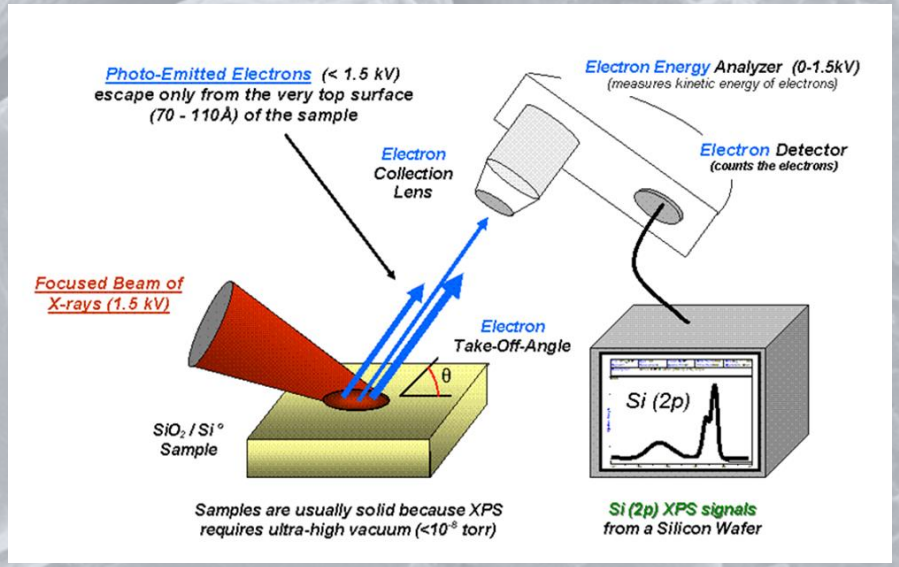
Principle of Operation

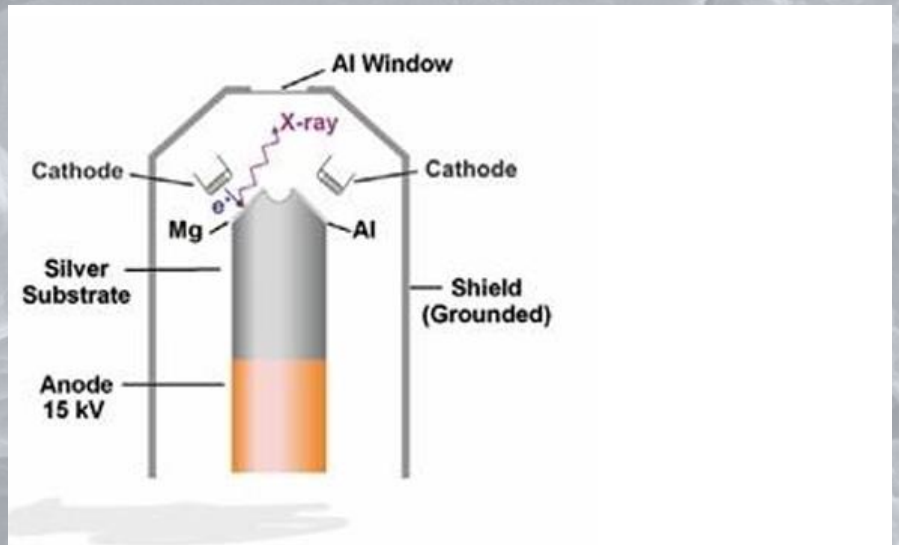
The X-ray tube generates the raw Al Kα radiation.

The crystal monochromator selects only the **$K\alpha_1$ component**, rejecting satellites and Bremsstrahlung.

The monochromator also **focuses** the X-ray beam onto the sample, forming a well-defined spot.

This yields a photon beam of extremely narrow energy bandwidth, enabling precise binding-energy measurement.





ANODE	RADIATION	ENERGY (EV)	LINE WIDTH (EV)
Mg	Kα	1253.6	0.7
Al	Kα	1486.8	0.85
Zr	Lα	2042.4	1.6
Ag	Lα	2984.3	2.6

Main Characteristics

High energy resolution: Narrow X-ray bandwidth produces sharper, more symmetric peaks, greatly improving chemical-state sensitivity.

Reduced background noise: Removal of overlapping signals reduces spectral interference and strengthens signal-to-noise ratio.

Possibility of small spot sizes: Focused monochromatic beams can reach tens of micrometers, enabling micro-XPS or spatially resolved surface analysis.

More heat load and potential for damage: Focused high-intensity beams may locally heat or damage sensitive materials (polymers, organic films, biological samples).

Higher cost and more complex design: Requires precision optics, alignment, and cooling systems





۲.Ultra-High Vacuum (UHV) Chamber

The Ultra-High Vacuum (UHV) chamber is the central environment where XPS measurements are performed. XPS requires extremely low pressures, typically in the range of P<1×10−9 mbar

This pressure ensures that emitted photoelectrons can travel from the sample surface to the analyzer **without being scattered** by residual gas molecules.

Key Functions of the UHV Chamber

Maintains electron mean free path: In UHV, the electron mean free path becomes sufficiently long to allow accurate energy analysis.

Protects sample surface from contamination: Even a thin layer of adsorbed gas (e.g., oxygen, water, hydrocarbons) can alter the XPS signal. UHV minimizes adsorption during the experiment.

Provides a stable environment for X-ray optics and electron analyzers: Components such as the monochromator, electron analyzer, detector, and sample stage are aligned and operate reliably only under UHV conditions.

Reduces chemical reactions at the surface: Reactive species are nearly absent at UHV pressures, preventing oxidation or degradation during measurement.

3.Vacuum Pumps

Achieving and maintaining UHV conditions requires a **combination of pumping systems**, each responsible for a different pressure range. Modern XPS systems typically employ a sequence of pumps that gradually reduce chamber pressure from atmospheric levels to UHV.

1. Roughing and Medium-Vacuum Pumps

These pumps reduce pressure from atmospheric pressure to approximately 10−3–10−2 mbar

Common types include:

Rotary vane pumps

Scroll (dry) pumps

They remove large volumes of air and prepare the chamber for high-vacuum pumps.

2. High-Vacuum Pumps

Once the pressure reaches the 10−3mbar range, high-vacuum pumps take over:

Turbomolecular pumps (TMPs)

TMPs use rapidly spinning blades to remove gas molecules and can reach pressures down to approximately 10−7 mbar

3. UHV Pumps

To reach true UHV conditions required for XPS, additional pumps are used:

• Ion pumps: Use electric and magnetic fields to ionize and trap residual gas.

• Titanium sublimation pumps (TSPs): Coat the chamber interior with getter titanium, which chemically binds reactive gases.

• Non-evaporable getter (NEG) pumps: Absorb active gases without heating.





۴.Sample Stage / Manipulator

The sample stage (also known as the manipulator) positions the specimen within the XPS chamber and allows precise control over its orientation and environment. Correct sample positioning is crucial because XPS analysis is highly surface-sensitive.

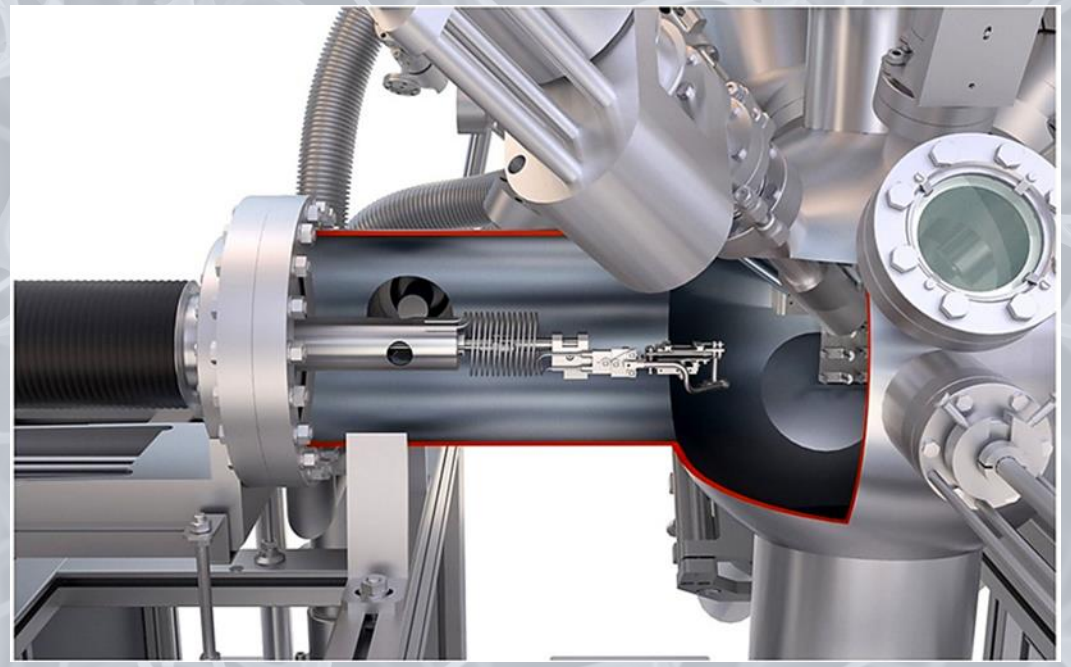
Key Capabilities

Multi-axis movement: Typical manipulators allow 3 or 5 degrees of freedom (x, y, z) and often θ, ϕ enabling precise alignment with the X-ray beam and electron analyzer.

Heating and cooling: Many stages support temperature control, often from cryogenic levels (≈ 100 K) up to high temperatures (> 1000 K), enabling in-situ studies of reactions, diffusion, or surface treatments

Electrical isolation:

For insulating samples, charge compensation systems may be integrated into the stage to reduce surface charging





5.Electron Energy Analyzer (EEA)

The **Electron Energy Analyzer (EEA)** is a critical component in XPS that measures the kinetic energy (Ek) of photoelectrons emitted from the sample surface. The binding energy (Eb) of an electron is then calculated using the equation:

$$Eb = h\nu - Ek - \phi$$

where:

$h\nu$= photon energy of the X-ray source

Ek= kinetic energy of the detected electron

ϕ= work function of the spectrometer

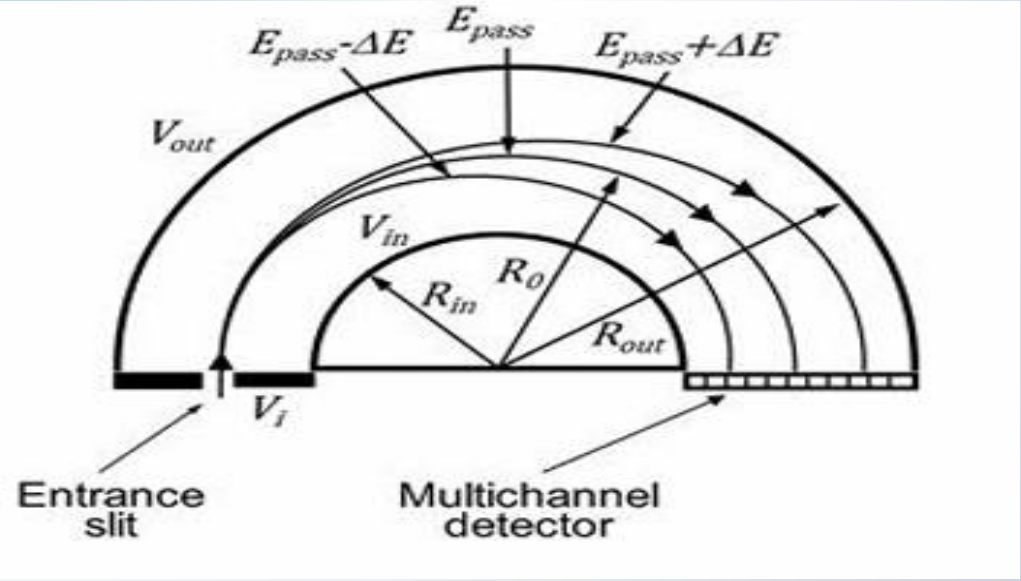
Accurate measurement of Ek is essential because the **chemical state information** depends directly on small shifts in binding energy.

1. Hemispherical Electron Analyzer (HSA)

There are many different designs of electron energy analyzers, but the preferred choice for photoemission experiments, such as X-ray photoelectron spectroscopy (XPS), is the hemispherical analyzer (HSA). It uses a radial electric field between two hemispherical surfaces to disperse the electrons according to their kinetic energy, providing high resolution and sensitivity. Its operation is based on

the physical law of balancing the electrostatic force and the centrifugal force

The HSA consists of a pair of concentric hemispherical metals, between which there is a gap for the electrons to pass. The inner hemisphere has radius $R1$, and the outer hemisphere has radius $R2$ (with $R2 > R1$). There is usually a series of electrostatic



Potentials $V1$(inner hemisphere) and $V2$ (outer hemisphere) are applied such that $V1>V2$ (inner more positive than outer, creating a radially inward electric field for electrons). The applied potentials create an intermediate equipotential surface with mean radius $R0=(R1+R2)/2$, and the kinetic energy corresponding to electrons that follow a circular trajectory along this surface is called the pass energy (Ep, typically in eV).

If Ekinetic =Ep:

The two forces cancel out, and the electron maintains the circular path $R0$.

If Ekinetic >Ep (fast):

The centrifugal force is stronger. The electron is deflected too far outward and strikes the outer hemisphere wall.

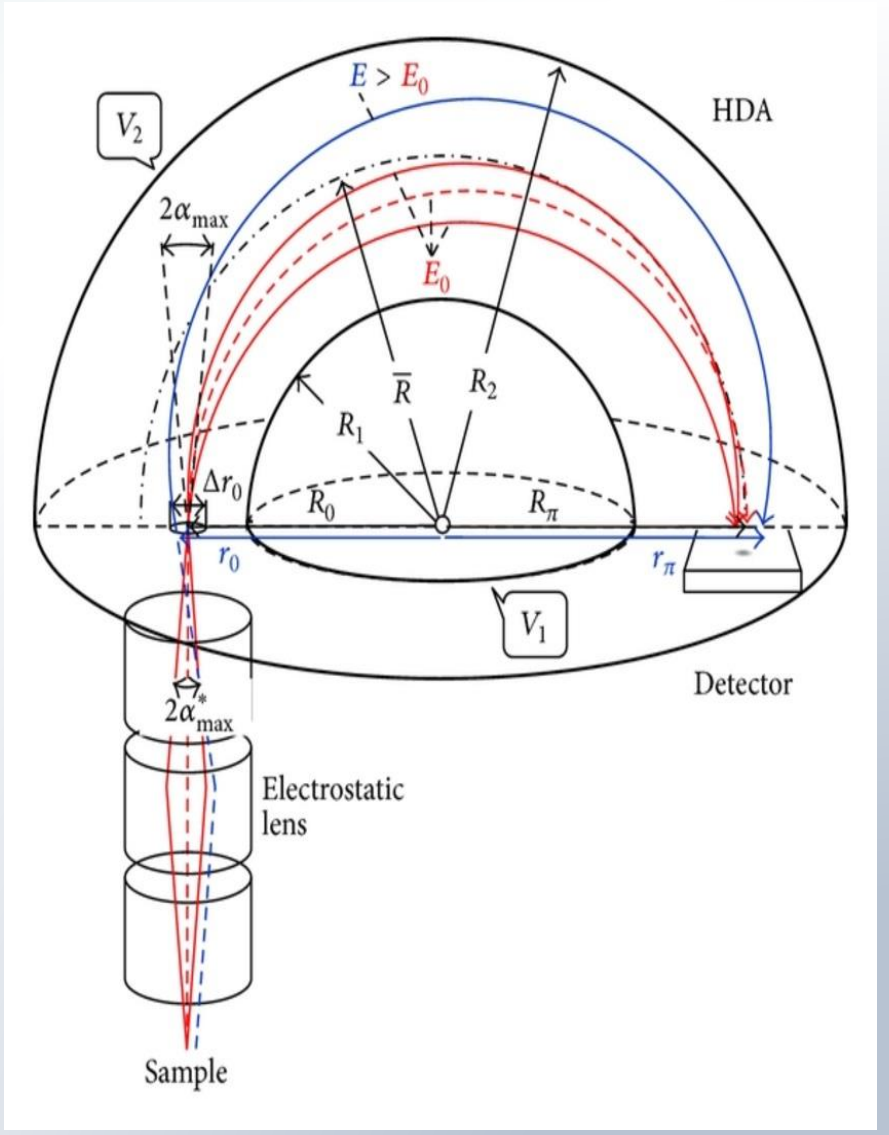
If Ekinetic <Ep (slow):

The electric force is stronger. The electron is attracted too strongly inward and strikes the inner hemisphere wall.

Since $R1$, $R2$, and R0 are constant, in principle $V1$ and $V2$are the parameters that are varied to set the pass energy. The potential difference is given by:

$$V1-V2=(Ep/e)(R2/R1-R1/R2)$$

where e is the elementary charge



lenses between the sample and the analyzer. The kinetic energy of the electrons emitted from the sample is normally too high for the analyzer to achieve high energy resolution; therefore, their speed must be reduced. This deceleration is achieved within the lens system through a process called electron retardation

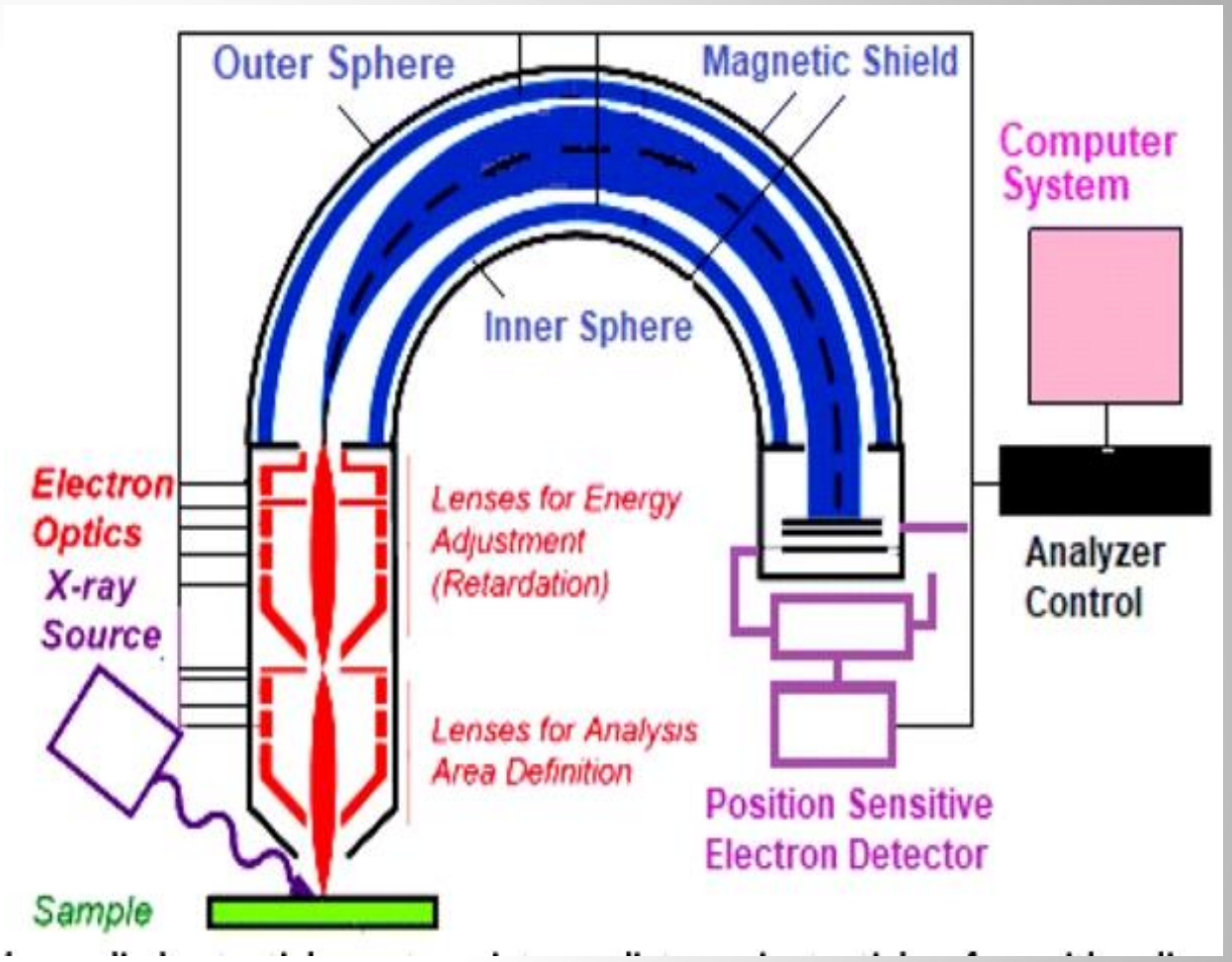
Before being injected into the HSA, the electrons are focused by electrostatic transfer lenses, and their energy is reduced to the pass energy level.

The multi-element electrostatic lens system:

1.Collects electrons from a large angular distribution.

2.Focuses the electrons onto the entrance aperture.

3.Retards the electrons to the pass energy.



The energy resolution (ΔE) of the analyzer is a critical parameter and is approximated by:

$$\Delta E = E_p\left(\frac{w}{2R_0} + \alpha^2\right)$$

where:

ΔE= energy resolution (full width at half maximum, FWHM),

E_p= pass energy,

w= slit width (average of entrance and exit slits),

R_0= mean radius,

α= maximum acceptance angle (in radians).

This formula highlights that resolution improves with larger analyzer radius, narrower slits, smaller acceptance angles, and lower pass energies, though trade-offs exist with signal intensity and transmission efficiency.



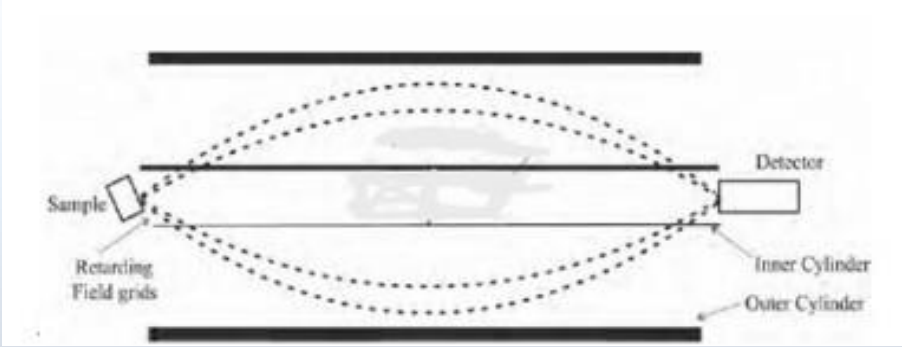
2. Cylindrical Mirror Analyzer (CMA)

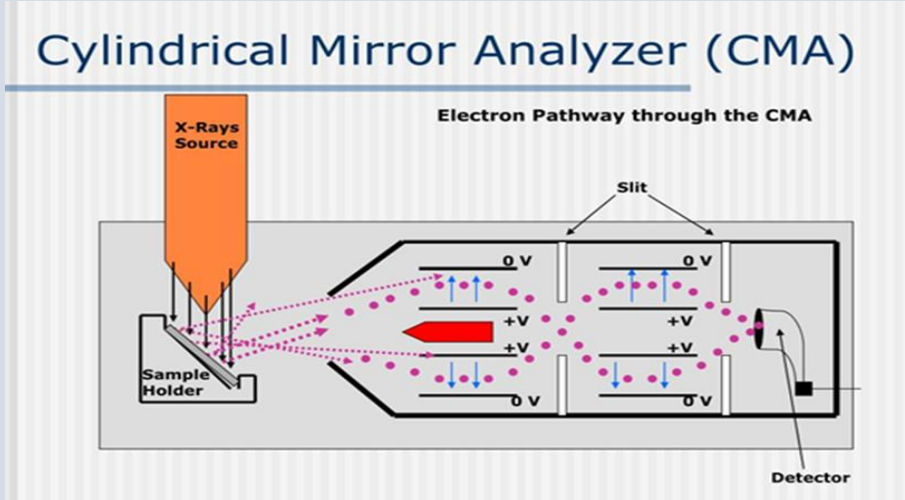
The Cylindrical Mirror Analyzer (CMA) is a type of electron energy analyzer historically important in surface analysis, particularly in Auger Electron Spectroscopy (AES), but less commonly used today in modern X-ray Photoelectron Spectroscopy (XPS) due to limitations in energy resolution and flexibility.

The CMA consists of two coaxial (concentric) cylinders:

- An inner cylinder, usually grounded (0 V).
- An outer cylinder, held at a negative potential ($-V_2$).

Electrons emitted from the sample enter the analyzer through a small entrance aperture (typically ~2 mm) in the inner cylinder. A strong radial electrostatic field is established between the two cylinders. This field deflects electrons into curved trajectories, and only those with a specific kinetic energy are focused through a second defining aperture onto the detector after completing a characteristic path at an angle of approximately 42.3° relative to the axis (the so-called “magic angle” for second-order focusing). The relationship between the kinetic energy of the transmitted electrons (E_k) and the deflection voltage is:





$$E_k = K \cdot | V_2 |$$

where:

- E_k is the kinetic energy of the electrons (in eV),
- V_2 is the absolute value of the negative voltage applied to the outer cylinder,
- K is a geometry-dependent constant (typically K ≈ 1.3 for most commercial single-pass CMAs).

6.Detectors in XPS

The most important detectors used in XPS are the **Position-Sensitive Detector (PSD)** and the **Channeltron Electron Multiplier**, while amplification is commonly performed by a **Microchannel Plate (MCP)**. Additional advanced detectors such as delay-line or pixel-based systems may be used in specialized instruments.

1. Microchannel Plate (MCP)

(Electron Multiplier – Not a Standalone Detector)

Although often mentioned together with detectors, the MCP itself is **not a complete detector**. It is an electron multiplier used to amplify very weak signals before they reach the actual detection element.

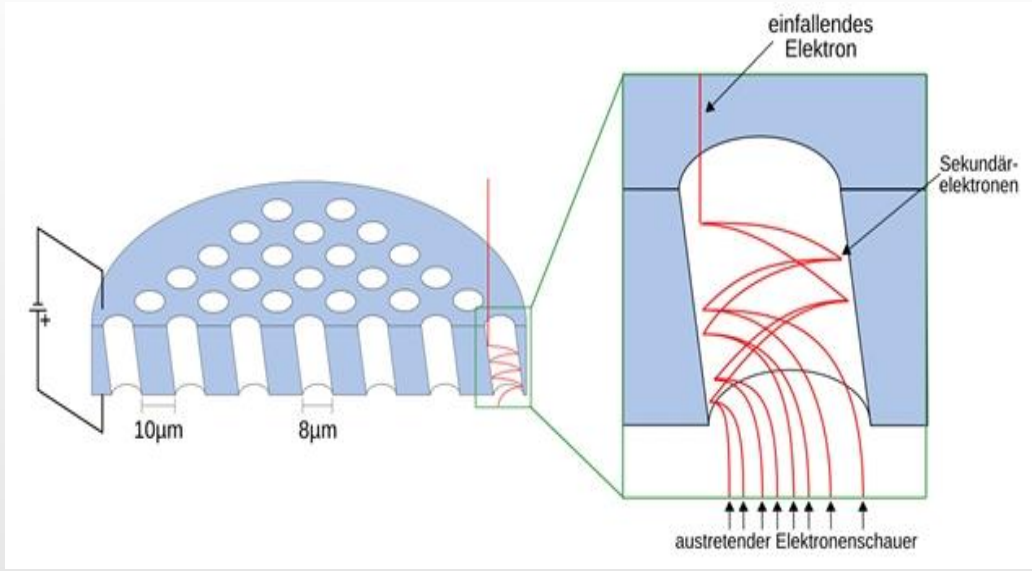
A microchannel plate is a thin (~0.5–1 mm) disk of lead-glass containing millions of tiny cylindrical channels (typically 4–25 µm diameter, spaced 6–30 µm apart) oriented at a small bias angle (5–15°) to the surface

Operating principle:

- A potential difference of ~800–1200 V is applied across the MCP.
- Each channel acts as an independent continuous-dynode electron multiplier.
- An incoming electron strikes the channel wall near the entrance and liberates secondary electrons.
- These secondary electrons are accelerated by the electric field along the channel, strike the opposite wall, and produce more secondaries.
- The cascade continues along the ~1 mm length of the channel, resulting in a gain of 10^3–10^4 electrons per incident electron.

Role in XPS

The MCP acts as the **first amplification stage**, ensuring that very weak electron signals can be detected by the final detector (PSD, CCD, delay-line detector, etc.).





https://torrscientific.co.uk/mcps-and-phosphor-screens/

2. Position-Sensitive Detector (PSD)

(Most Important Detector in Modern XPS)

The PSD is the primary detection technology used in nearly all high-performance hemispherical analyzers (e.g., Scienta Omicron, SPECS, Kratos). It can measure not only the intensity of the electrons but also their **position**, which corresponds to energy or angle

1. Micro-Channel Plates - MCPs

Photoelectrons emitted from the sample surface have a very weak signal. Before detection, they must be amplified.

2. Readout System

After amplification, the electron cloud must be position-determined. There are two main technologies for this purpose

a. Delay-Line Detector - DLD

This is the most precise and fastest type. The electron cloud strikes a wired grid. The time it takes for the signal to reach both ends of the wire is measured. The time difference accurately indicates where the electron has hit.

Advantage: Very high counting speed (millions of counts per second) and excellent resolution

b. Fluorescent screen and CCD/CMOS camera:

The electron cloud strikes a phosphor screen and produces light. A digital camera observes this light from behind and records its position

Advantages

- Very high count-rate capability
- Fast scanning and high throughput
- Excellent energy resolution when combined with high-quality analyzers

Limitations

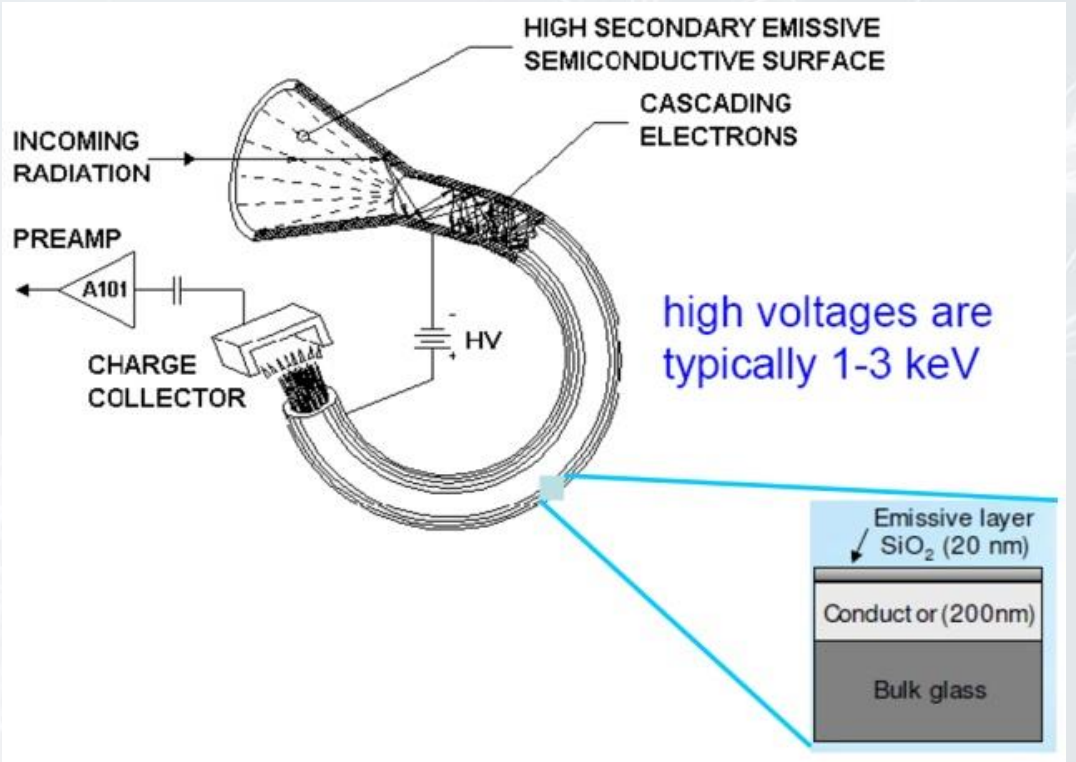
- More complex and expensive compared to single-channel detectors
- More expensive

3. Channeltron Electron Multiplier

(Legacy Detector – Still Used in Some XPS Systems)

A channeltron is a **single-channel electron multiplier** that converts electron impacts into measurable pulses. It was historically the standard detector in older XPS instruments and is still used for applications requiring simple intensity measurements.

1. **Primary Electron Entry**: An electron emitted from the sample enters the narrow glass channel (tube-shaped) whose inner surface is conductive.
2. **Impact and Secondary Electron Generation**: When the primary electron strikes the channel wall, several secondary electrons are released. This process is like a “cascade collision.”
3. **Acceleration by Electric Field**: A negative voltage is applied along the channel. This electrostatic field accelerates the secondary electrons toward the end of the channel, causing them to hit the wall again.
4. **Electron Cascade** :Each new impact produces more electrons. This cycle repeats multiple times, and the number of electrons increases exponentially.
5. **Amplified Pulse Output** :At the end of the channel, a strong electron pulse (much larger than the initial signal) is collected. This pulse is then sent to the detection electronics and recorded as a measurable signal



7.The control and data storage system performs three main tasks:

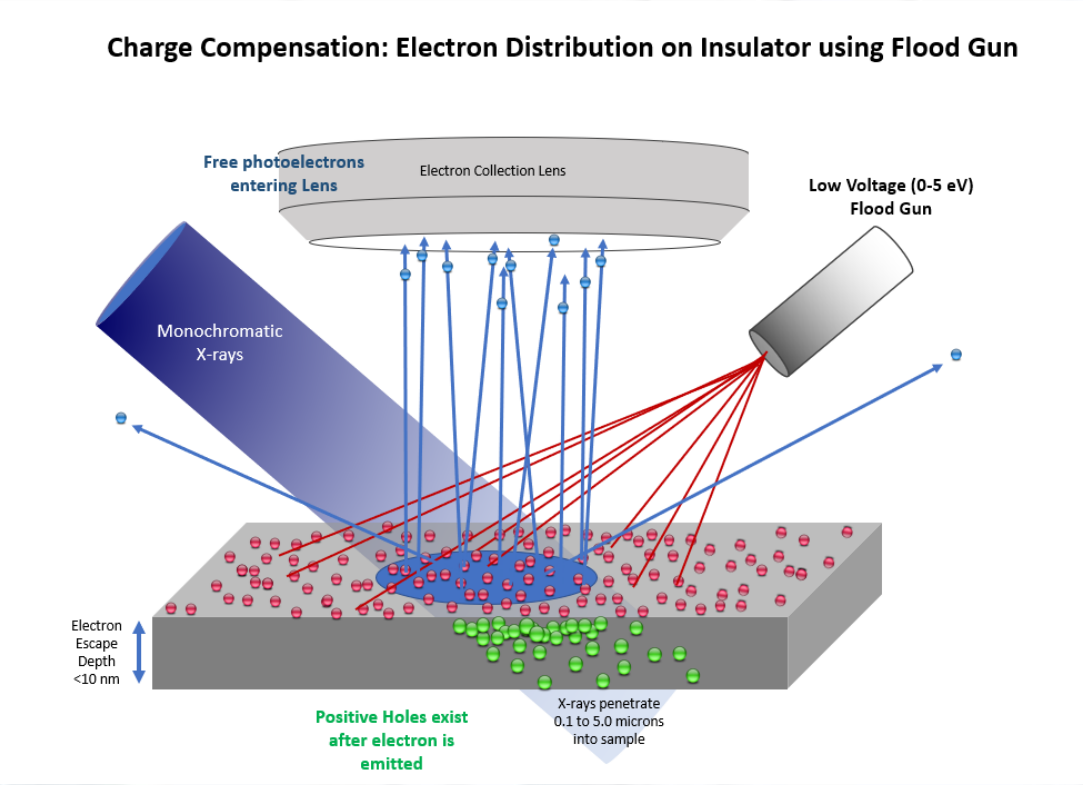
1.Controls the entire XPS instrument — it regulates voltages, currents, photon energy, vacuum pumps, the energy analyzer, and detectors, ensuring all components work in coordination.

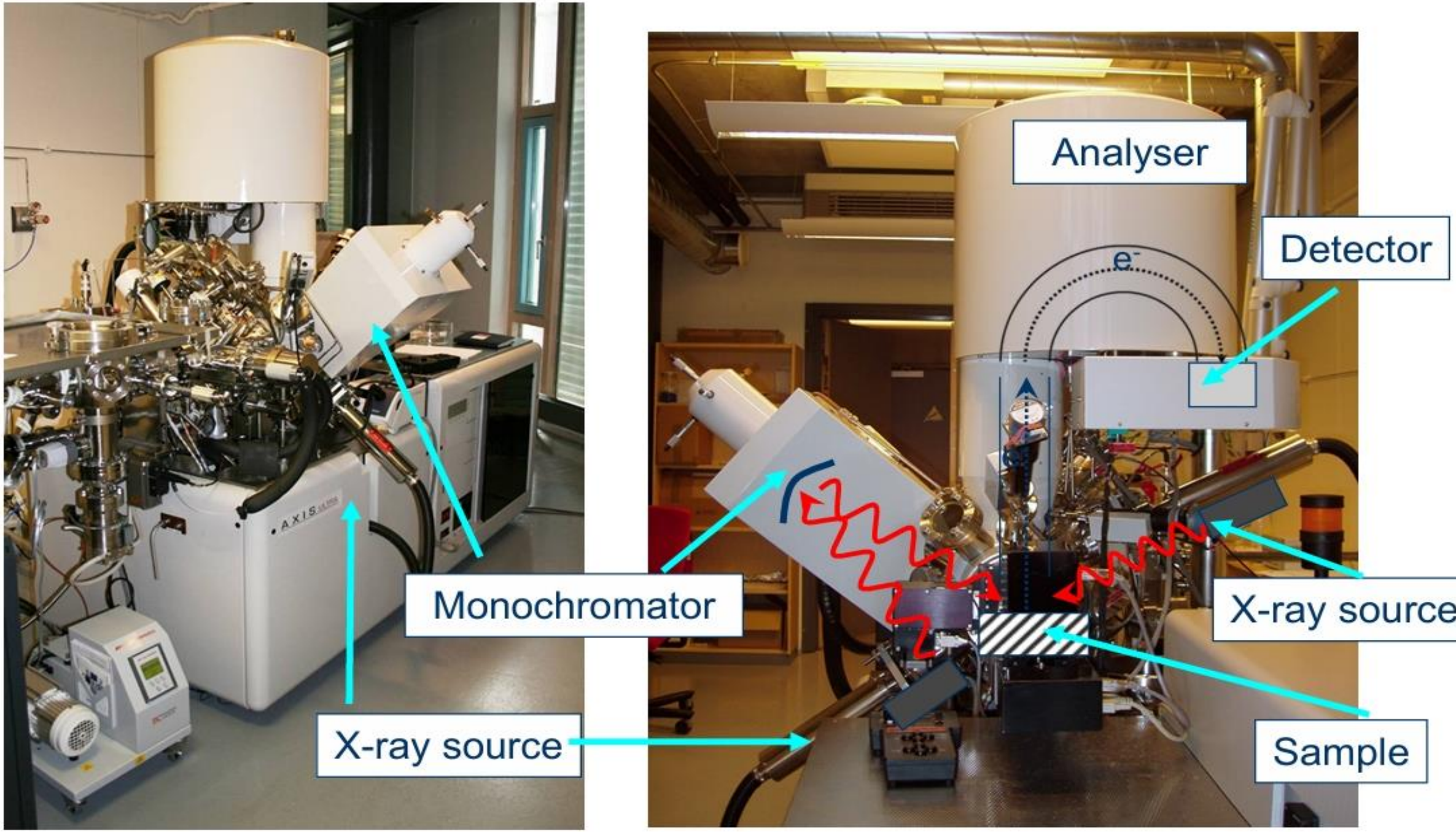
2.Receives and processes the electron signal — it takes the raw signal from detectors (such as MCP or PSD) and converts it into readable information: electron intensity and energy spectra.

3.Stores and displays data — it saves XPS spectra (binding energy vs. intensity) and provides tools for processing, peak fitting, noise reduction, and surface chemical analysis.

Charge Neutralization System (Flood Gun / Electron Neutralizer)

Insulating and poorly conductive samples accumulate positive charge during XPS measurements because photoemission removes electrons from the surface. This charging leads to undesirable shifts in binding energy, peak broadening, and in severe cases, complete loss of spectral resolution. To correct this problem, modern XPS instruments use a **charge neutralization system**, commonly referred to as a **flood gun** or **electron flood source**




Analyser
Detector
e-
Monochromator
X-ray source
X-ray source
Sample

Understanding XPS Spectra

1. Elemental Specificity

What elements are present in the sample?

Each chemical element has a **unique set of core-level binding energies**. These binding energies serve as a "fingerprint," enabling direct identification of surface elements.

Peak Identification

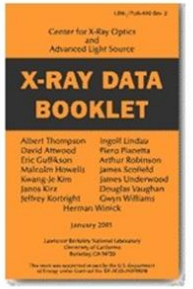
Identification is performed by comparing the measured binding energies to well-established reference databases.
Since different chemical states of the same element slightly shift the binding energy, XPS can distinguish between:

- Elements (e.g., C, O, N, Si, Fe, Cu, etc.)
- Chemical states (e.g., metallic Fe vs. FeO vs. Fe_2O_3)
- Functional groups (e.g., C–C vs. C–O vs. C=O)

Binding energies are intrinsic properties and do not depend on the photon energy used to measure them.

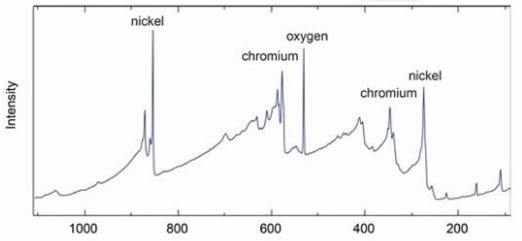
Table 1-1. Electron binding energies, in electron volts, for the elements in their natural forms.

Element	K 1s	L_1 2s	L_2 $2p_{1/2}$	L_3 $2p_{3/2}$	M_1 3s	M_2 $3p_{1/2}$	M_3 $3p_{3/2}$	M_4 $3d_{3/2}$	M_5 $3d_{5/2}$	N_1 4s	N_2 $4p_{1/2}$	N_3 $4p_{3/2}$
1 H	13.6											
2 He	24.6*											
3 Li	54.7*											
4 Be	111.5*											
5 B	188*											
6 C	284.2*											
7 N	409.9*	37.3*										
8 O	543.1*	41.6*										
9 F	696.7*											
10 Ne	870.2*	48.5*	21.7*	21.6*								
11 Na	1070.8†	63.5†	30.65	30.81								
12 Mg	1303.0†	88.7	49.78	49.50								
13 Al	1559.6	117.8	72.95	72.55								
14 Si	1839	149.7*b	99.82	99.42								
15 P	2145.5	189*	136*	135*								
16 S	2472	230.9	163.6*	162.5*								
17 Cl	2822.4	270*	202*	200*								
18 Ar	3205.9*	326.3*	250.6†	248.4*	29.3*	15.9*	15.7*					
19 K	3608.4*	378.6*	297.3*	294.6*	34.8*	18.3*	18.3*					
20 Ca	4038.5*	438.4†	349.7†	346.2†	44.3 †	25.4†	25.4†					
21 Sc	4492	498.0*	403.6*	398.7*	51.1*	28.3*	28.3*					
22 Ti	4966	560.9†	460.2†	453.8†	58.7†	32.6†	32.6†					

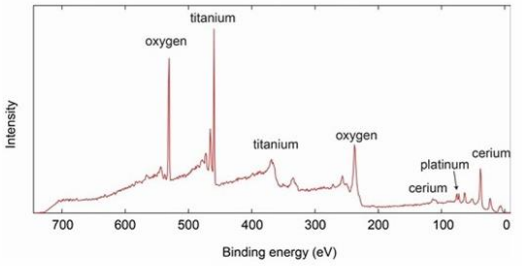


Elemental Specificity

Nickel-Chromium alloy –
corrosion resistant material



$Pt/CeO_2/TiO_2$
catalyst for methanol production

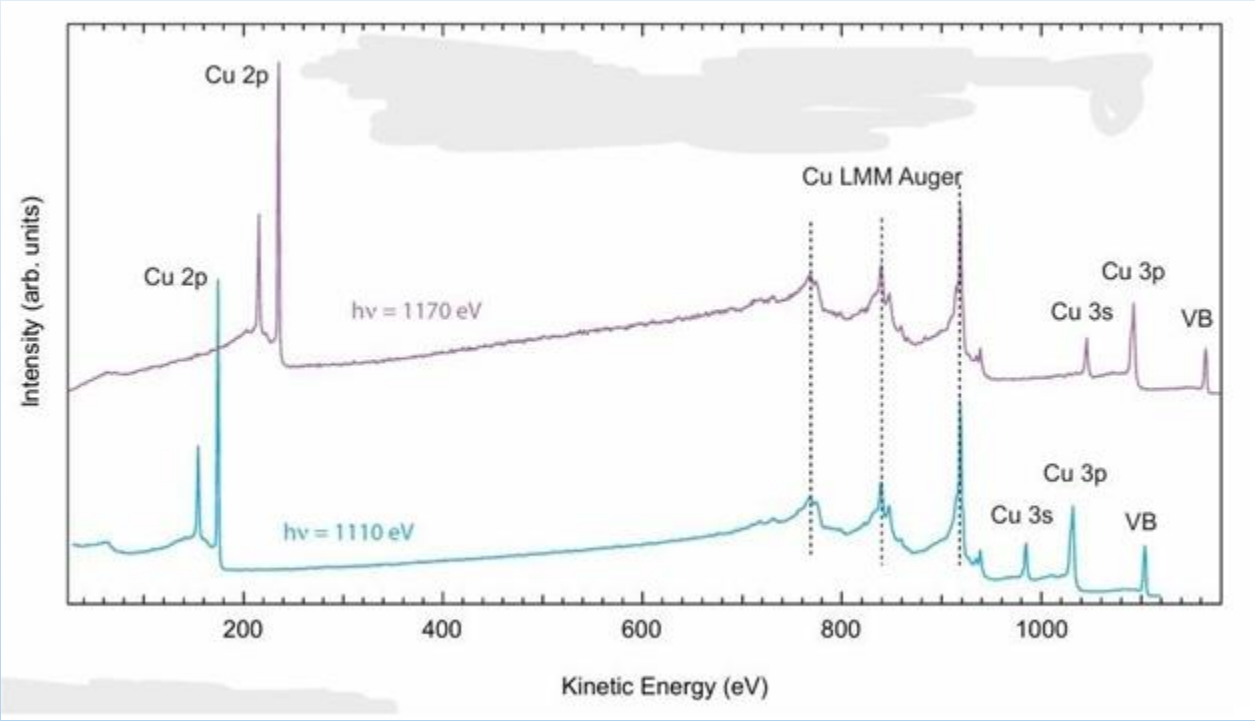


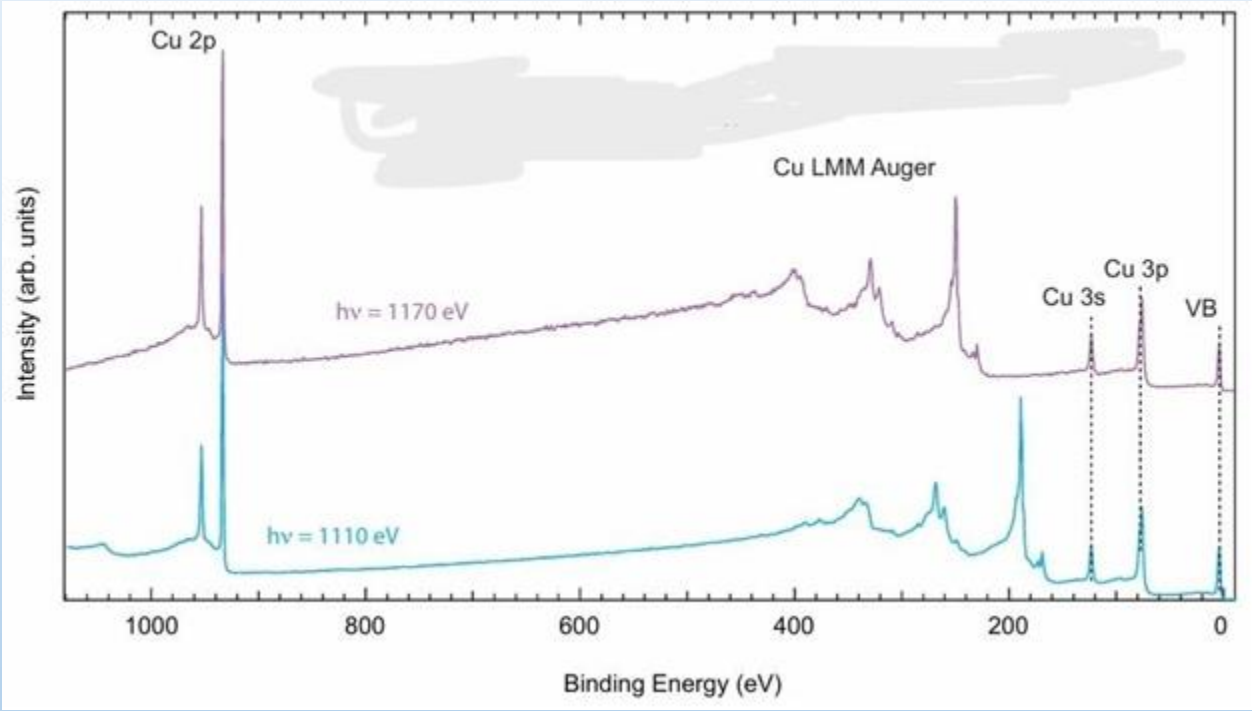
2. Differentiating XPS Peaks from Auger Peaks

XPS spectra typically include both:

Photoelectron peaks (true XPS peaks — binding-energy dependent)

Auger electron peaks (secondary electrons — photon-energy independent)



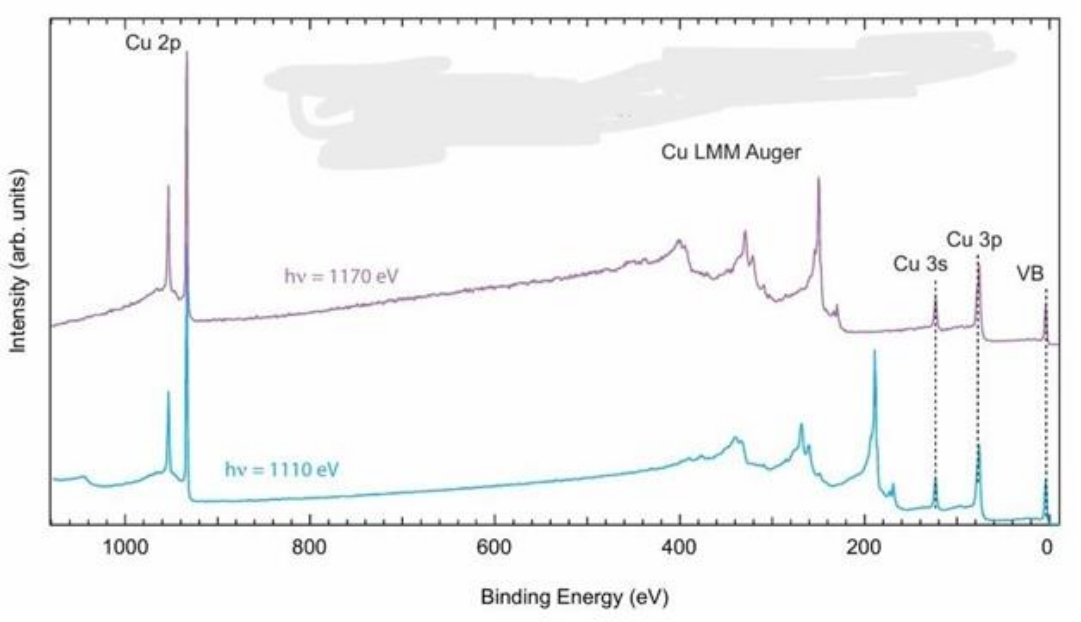


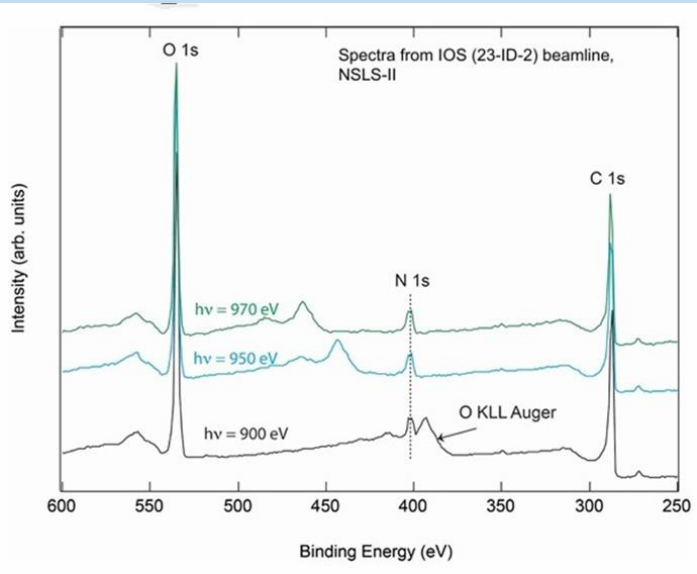
Change photon energy:
XPS kinetic energy changes
Auger kinetic energy stays the same

$Eb=h\nu-Ek-\phi$

Change photon energy:
XPS binding energy is constant
Auger peaks move

2. Differentiating XPS Peaks from Auger Peaks

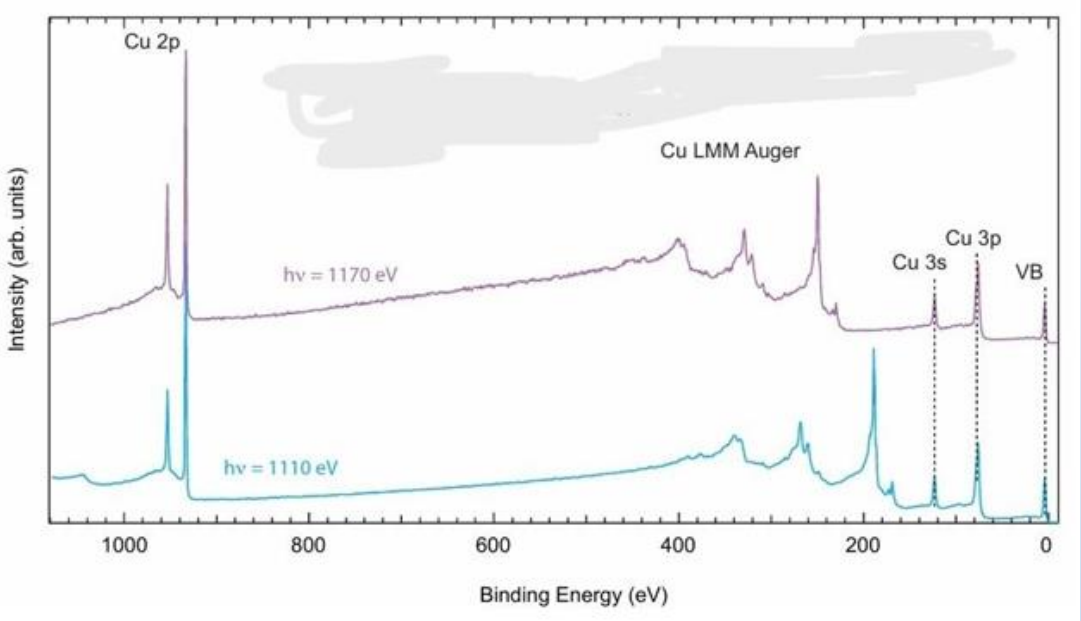




Key Distinguishing Features

Feature	XPS Photoelectron Peaks	Auger Electron Peaks
Dependence on photon energy	**Kinetic energy changes** when X-ray energy changes	**Kinetic energy stays constant**
Binding energy	Constant and well-defined	Not directly used for chemical analysis
Peak shape	Sharper, symmetric (FWHM ~0.8–1.5 eV)	Broader, asymmetric (FWHM ~5–10 eV)
Behavior when X-ray energy changes	XPS peaks **shift** in kinetic energy	Auger peaks **move** in binding-energy representation

2. Spin-Orbit Splitting and Branching Ratios



When an electron is emitted from an orbital such as 2p, the energy level splits into two sub-levels, $2p_{1/2}$ and $2p_{3/2}$, due to spin-orbit coupling.
Intrinsic Intensity Ratio (Branching Ratio): According to quantum mechanics, the intensity of these peaks is proportional to the number of occupied states, defined by (2j+1)

For p orbitals (where j values are 1/2 and 3/2): The intensity ratio is always 1:2.
For d orbitals: The intensity ratio is 2:3
For f orbitals: The intensity ratio is 3:4

Why is this a "fingerprint"? These ratios remain constant for every atom, regardless of its chemical compound. If the area ratio of the smaller peak to the larger peak in the copper spectrum (Cu 2p) shown in the image is not exactly 1:2, it indicates the presence of overlapping peaks from other elements or features like shake-up satellites.

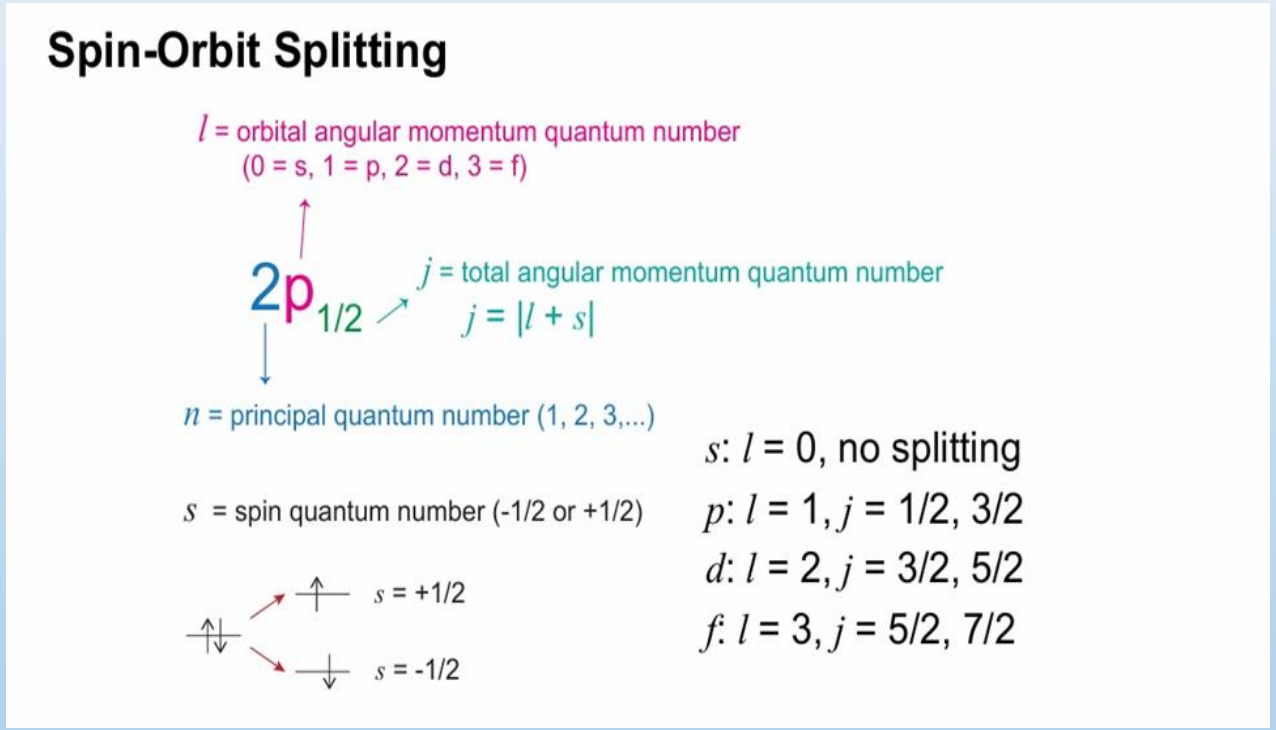
The "Fingerprint" Application:
Any deviation from these stoichiometric ratios or the expected spin-orbit splitting (ΔE) indicates:
1. Chemical state changes (e.g., Shake-up satellites in Cu^{2+})
2. Peak overlap from contaminating elements
3 . Multiplet splitting

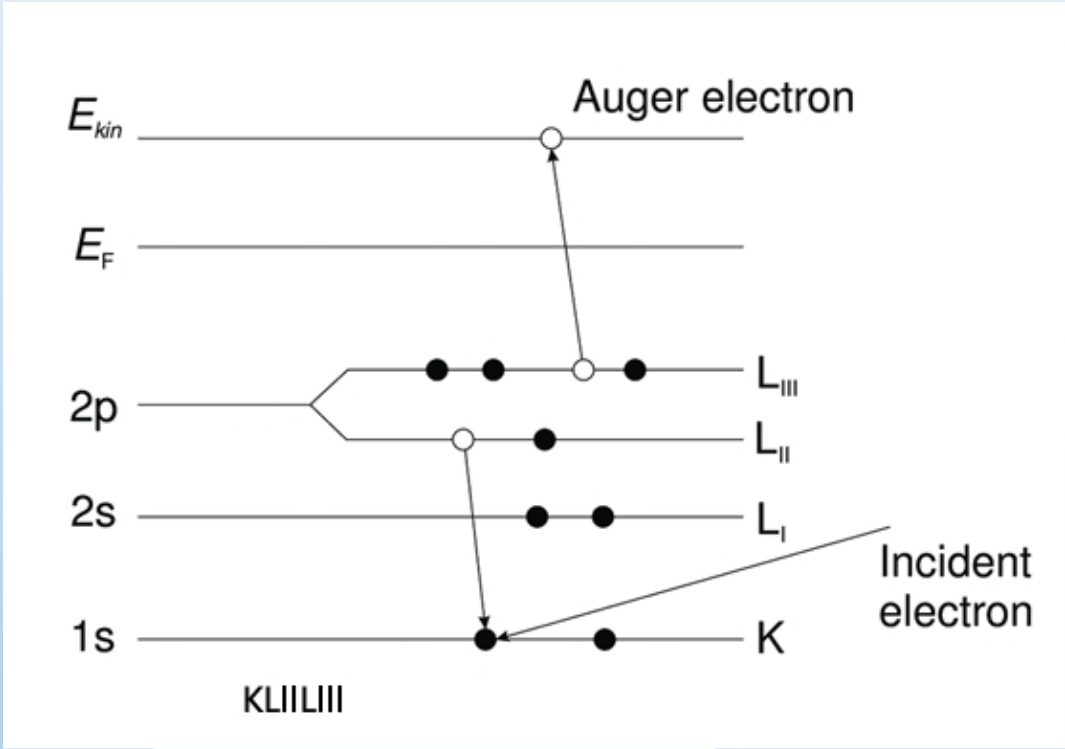
3. Spin–Orbit Splitting

Most core-levels of heavier elements are split into two components due to spin–orbit coupling. For example:

p orbitals → $p_{1/2}$ and $p_{3/2}$

d orbitals → $d_{3/2}$ and $d_{5/2}$





4. Chemical Sensitivity (Chemical Shifts)

What is the chemical environment of each element?

XPS is capable of determining how atoms are bonded because the **binding energy is highly sensitive** to the local chemical environment.

Why chemical shifts occur

When an atom is bonded to a more electronegative element (e.g., O, N, F), electron density around that atom decreases.
As a result:

Higher oxidation state ⇒ Higher binding energy

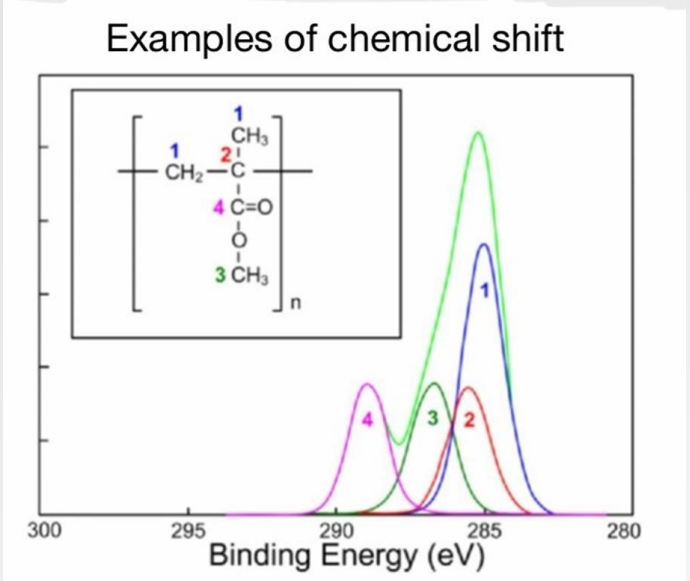
Example — carbon bonding environments:

- C–C → ~284.8 eV
- C–O → ~286.5 eV
- C=O → ~288.5 eV

What chemical shifts reveal

- Oxidation states (e.g., Fe^{2+} vs. Fe^{3+})
- Types of chemical bonds (C–C, C–O, C=O, Si–O, Si–C, etc.)
- Presence of functional groups in organic materials

Chemical sensitivity is one of the most powerful strengths of XPS.



When an atom is bonded to more electronegative elements (e.g., O, F, N), electron density is withdrawn, increasing the core-level binding energy. Conversely, bonding to less electronegative or metallic environments decreases the binding energy.

These chemical shifts are the foundation for identifying functional groups, oxidation states, coordination chemistry,etc

Typical chemical shift ranges (relative to elemental or hydrocarbon reference):

Element / Bond type	Binding Energy Range (eV)	Typical Reference Value
C 1s (C–C, C–H, adventitious carbon)	284.5 – 285.2	284.8 eV
C 1s (C–O, alcohols, ethers)	286.0 – 286.8	
C 1s (C=O, carbonyls)	287.5 – 288.5	
C 1s (O–C=O, carboxylates, esters)	288.8 – 289.5	
C 1s (carbonates, CF_2, etc.)	> 290	
O 1s (metal oxides)	529.5 – 531.0	
O 1s (organic oxygen, –OH)	532.0 – 533.5	
Si 2p (elemental Si)	99.0 – 99.5	
Si 2p (SiO_2)	103.0 – 103.8	
N 1s (amines)	~399.5	
N 1s (nitrides)	~397.0	
Fe $2p_{3/2}$ (metallic Fe)	706.7 – 707.1	
Fe $2p_{3/2}$ (Fe^{2+})	709.0 – 710.0	
Fe $2p_{3/2}$ (Fe^{3+})	710.5 – 712.0	

Depth Profiling (Sputter Depth Profiling)

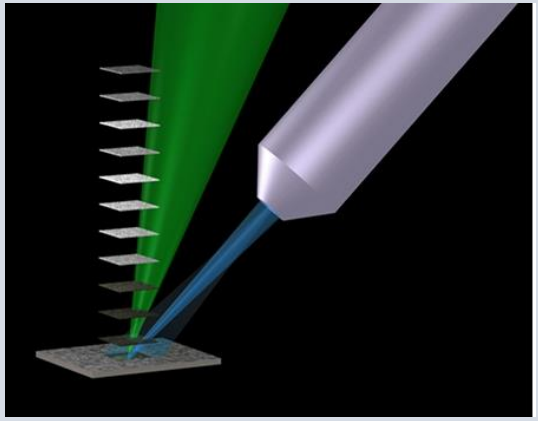
Depth profiling in XPS is a destructive technique that reveals the chemical composition as a function of depth by alternately removing thin layers of material using ion sputtering and acquiring XPS spectra after each cycle.

1) Principle and Workflow : An ion gun (usually Ar^+) bombards the surface with energetic ions (typically 0.5 – 5 keV).

2) Momentum transfer causes a collision cascade, ejecting surface atoms (physical sputtering).

3) Material removal rate: typically 0.1 – 10 nm/min, depending on ion energy, current density, material, and incident angle.

4) Typical cycle: sputter for a defined time → stop ion beam → acquire XPS spectrum → repeat.



Limitations

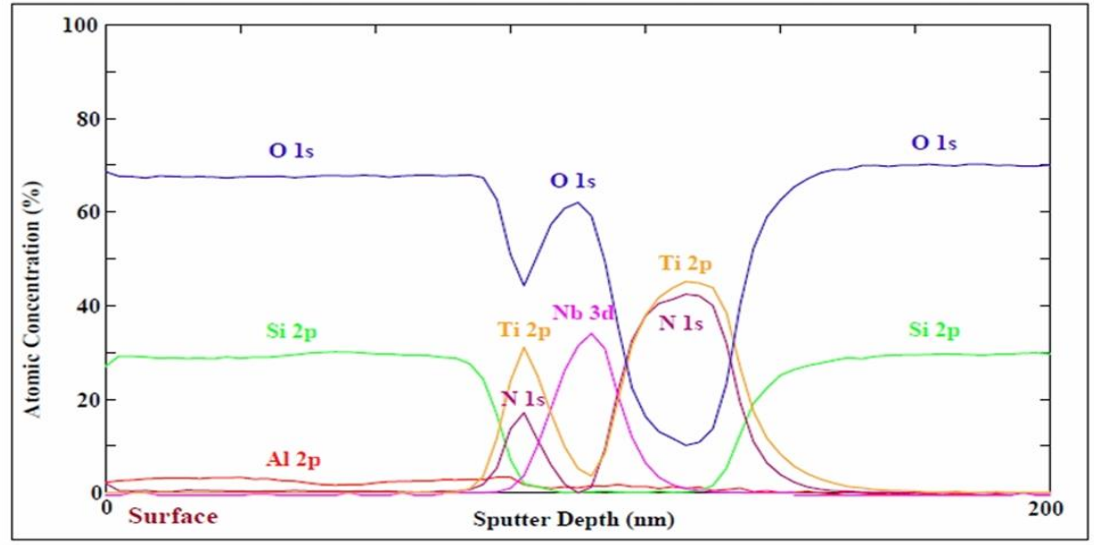
Destructive and irreversible

Chemical state information can be altered (reduction of oxides, bond breaking in organics)

Not suitable for polymers or beam-sensitive materials (unless low-energy or cluster ions are used)

Ion sputtering causes chemical changes on the surface (for example, it can break bonds or change oxidation states) and can cause surface roughness

Example



Atomic concentration for the elements of Glass Coating sample

Quantitative Analysis

XPS is a quantitative surface analysis technique. The intensity (peak area) of a photoelectron peak is directly proportional to the number of atoms contributing to that signal.

Basic Quantification Equation

$$C_i = \frac{I_i/S_i}{\sum_j (I_j/S_j)} \times 100\%$$

where:

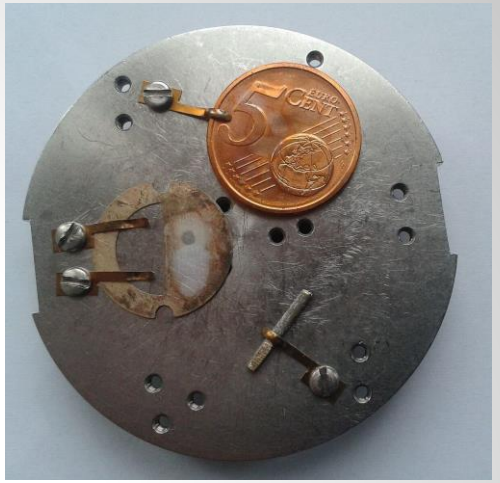
- C_i= atomic concentration of element i (at.%)
- I_i= measured peak area of element i (after background subtraction)
- S_i= relative sensitivity factor (RSF) for the corresponding photoelectron line
- The sum is taken over all detected elements (including carbon and oxygen)

Important Considerations

- Use consistent background type (Shirley or Tougaard) across all peaks.
- Detection limit: typically 0.1 – 1 at.% (depends on element and acquisition time).
- Accuracy: usually ±5–10% for homogeneous samples, worse for rough or inhomogeneous surfaces.

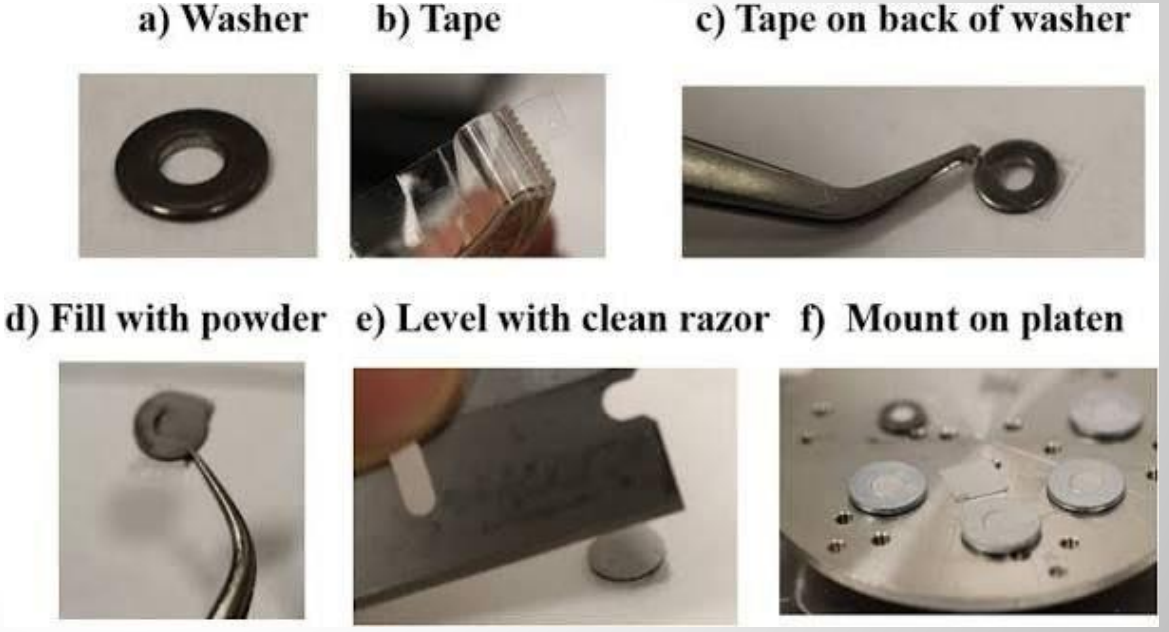
Sample Preparation and Handling for XPS Analysis

Proper sample preparation is crucial for obtaining accurate, reproducible, and artifact-free XPS data. Contamination, charging, and geometry effects can easily dominate or obscure the true surface chemistry if samples are not handled correctly.



1. General Requirements

- Sample size
- Surface
- Never touch the analysis area with bare hands



2. Cleaning Procedures

- Solvent cleaning (most common)
- In-situ cleaning (inside the XPS system)

3. Powder Samples

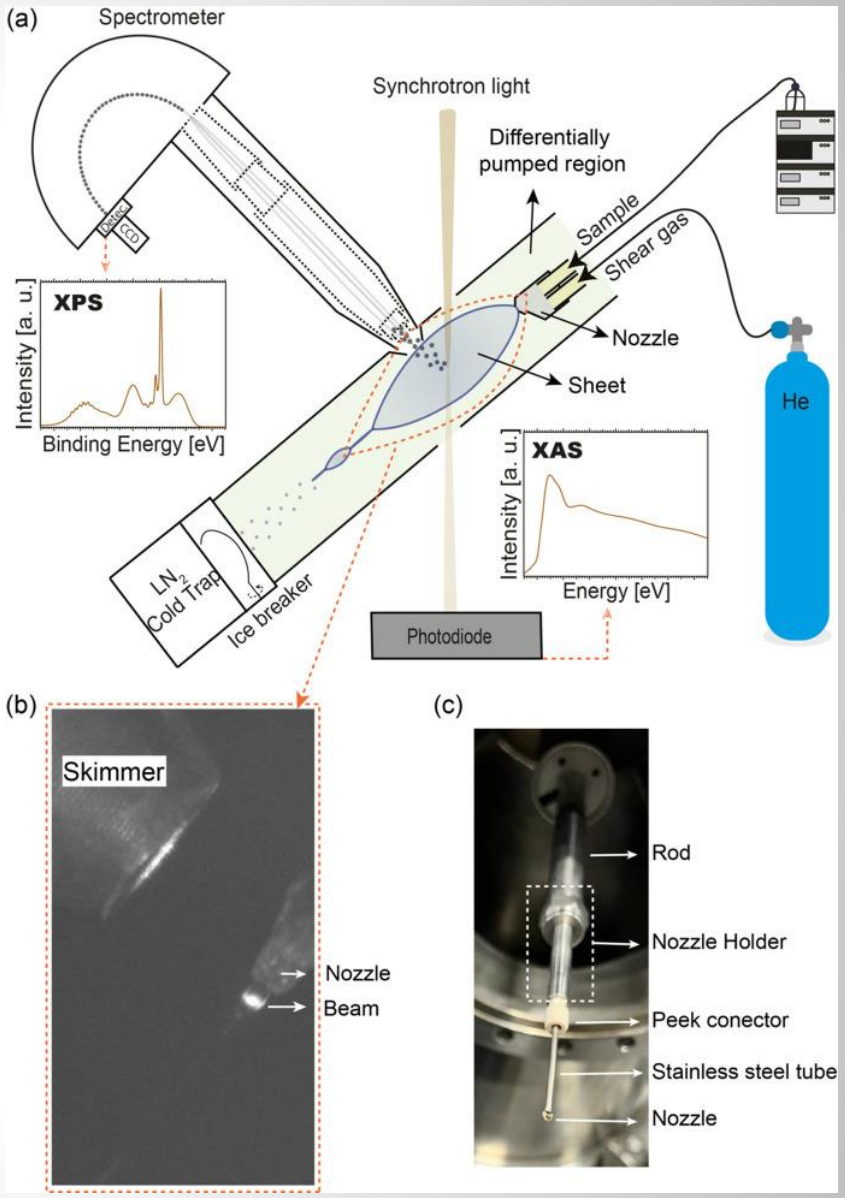
4. Liquid and Volatile Samples

Liquid and Gas Samples Analyzing liquid and gas samples with XPS is difficult and usually requires specialized equipment or a change in the physical state.

The Difficulty: To perform XPS analysis, the sample must be placed in an Ultra High Vacuum (UHV). Liquids and gases evaporate or sublimate in high vacuum, which not only makes analysis impossible but also contaminates the instrument.

Specialized Methods (for liquids):

- **Freezing:** The liquid can be analyzed while frozen at very low temperatures (such as liquid nitrogen temperature) in a vacuum.
- **Using a Flow Cell:** In some advanced setups, the liquid can be analyzed within a special chamber or flow cell (Jet Cell) under semi-vacuum conditions.



Advantages of X-ray Photoelectron Spectroscopy (XPS)

#	Advantage	Typical Applications / Benefit
1	High Surface Sensitivity	Ideal for thin films, coatings, surface treatments, contamination, corrosion products, catalysts
2	Near-Universal Elemental Detection	Comprehensive surface composition analysis across metals, semiconductors, insulators, organics
3	Chemical-State / Bonding Information	Distinguishes Fe^{0} / Fe^{2+} / Fe^{3+}, C–C / C–O / C=O, Si^{0} / SiO_2, sulfides vs sulfates, etc.
4	Quantitative Analysis	Direct surface stoichiometry, thickness of overlayers, interface composition
5	Non-Destructive (under normal conditions)	Repeated measurements on the same area, in-situ studies, valuable/art samples
6	Depth Profiling Capability	Multilayer structures, oxidation layers

Limitations of X-ray Photoelectron Spectroscopy (XPS)

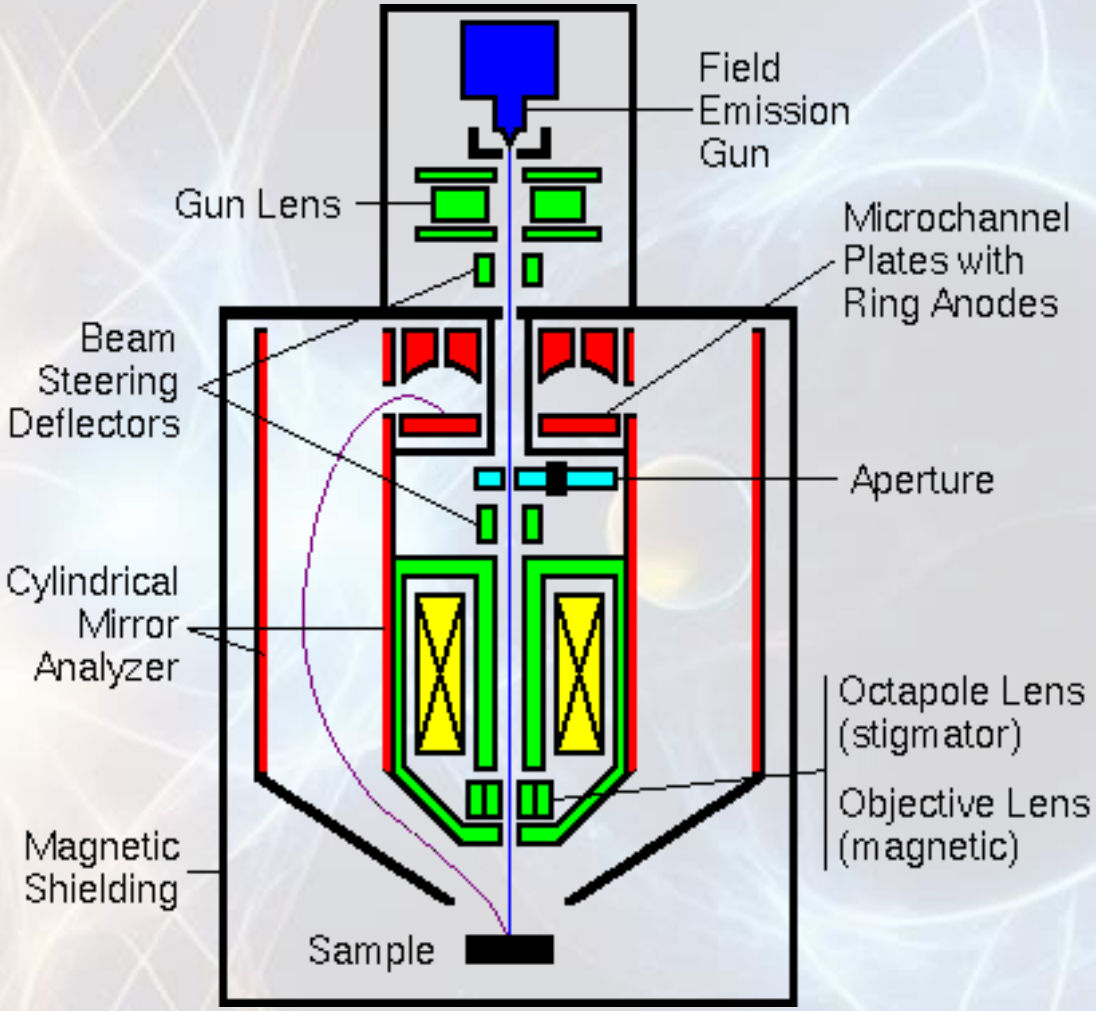
#	Limitation	Description and Consequences
1	Inability to Detect Hydrogen and Helium	H and He lack core electrons, producing no detectable photoelectrons.
2	Limited Sensitivity for Trace Elements	Detection limit remains ~0.1–1 at.% even with high-flux sources; not viable for ultra-trace (ppm/ppb) impurities.
3	Strict Ultra-High Vacuum Requirement	UHV (10^{-9}–10^{-10} mbar) excludes liquids, volatiles, and many in-situ environmental studies without specialized setups.
4	Surface Charging in Highly Insulating Samples	Extreme insulators (e.g., thick dielectrics) can exhibit persistent differential charging, distorting peaks despite neutralization.
5	Destructive Nature of Depth Profiling	Ion sputtering induces chemical artifacts (e.g., preferential removal of light elements, oxide reduction) and atomic mixing (~2–5 nm zone).
6	Beam-Induced Chemical Changes	Focused X-ray beams (>1 keV) can cause subtle bond breaking or desorption in beam-sensitive organics and halides over extended scans.
7	Limited Depth Resolution in Profiling	Interface smearing due to sputter-induced roughening and mixing; typical resolution 3–15 nm, degrading with depth.
8	Quantification Challenges for Layered/Rough Surfaces	RSF-based models assume homogeneity; overlayers or topography lead to 20–50% errors without corrections.
9	No Direct Structural or Crystallographic Information	Reveals chemistry but not atomic arrangement, defects, or long-range order.
10	Sample Outgassing and Compatibility Issues	High-vapor-pressure components contaminate the system, risking pump damage or signal artifacts.
11	Time-Intensive for Deep Profiling	Sputter rates ~0.1–5 nm/min limit practical depths to <500 nm without excessive run times (hours to days).
12	Vacuum-Induced Sample Alteration	UHV can desorb physisorbed species or induce phase changes in reactive materials during transfer/hold.

#	Application	Real-World Example
1	Surface Elemental Identification	Intel uses XPS to detect trace Na/K contaminants on Si wafers, preventing yield loss in 7-nm chips (PHI case studies, 2023).
2	Chemical State and Bonding Analysis	NASA's Perseverance team confirmed Fe^{3+}/Fe^{2+} ratios in Martian regolith simulants (Nature Geoscience, 2022).
3	Thin Film and Coating Characterization	Boeing evaluates anti-corrosion coatings on 787 Dreamliner aluminum alloys (Boeing R&T report, 2024).
4	Surface Contamination Detection	Samsung identifies organic residues on OLED substrates, reducing defects in Galaxy S-series (Thermo Fisher, 2023).
5	Depth Profiling and Layered Structures	Tesla uses XPS depth profiling on NMC-811 cathodes to detect Mn dissolution, optimizing EV range (J. Power Sources, 2024).
6	Corrosion and Oxidation Studies	ExxonMobil identifies Cr^{3+} enrichment in pipeline steel inhibitors (Kratos Analytical report, 2023).
7	Polymer and Organic Surface Analysis	DuPont verifies plasma-etched fluoropolymer surfaces in Gore-Tex fabrics (DuPont case study, 2022).
8	Nanomaterials and Catalysis Research	BASF confirms Ru^{4+} active sites in RuO_2 catalysts for green hydrogen, achieving 20% efficiency gain (ACS Catalysis, 2024).
9	Semiconductor and Electronics	TSMC detects Zr^{4+} segregation in HfO_2/ZrO_2 stacks in 3-nm chips to minimize leakage (IEEE Trans. Electron Devices, 2023).

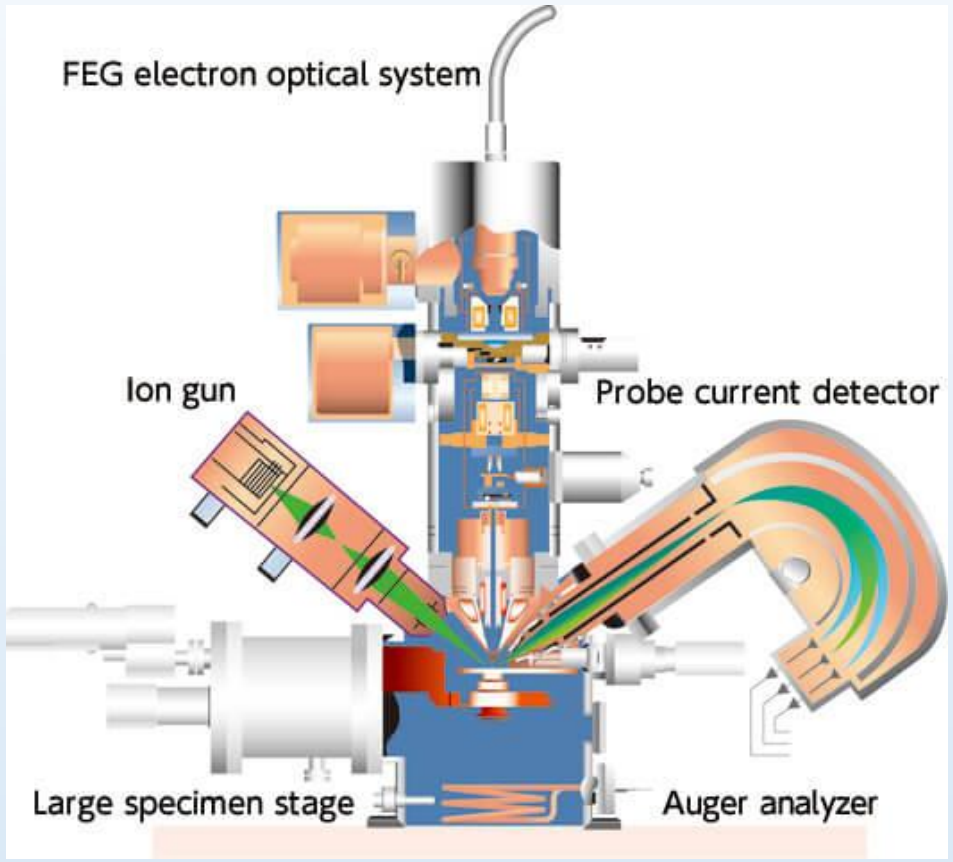
Main Competitors to XPS: A Brief Overview

X-ray Photoelectron Spectroscopy (XPS) excels in surface-sensitive chemical-state analysis, but faces competition from techniques like Auger Electron Spectroscopy (AES) and Energy-Dispersive X-ray Spectroscopy (EDS/EDX), which overlap in elemental detection but differ in resolution, depth, and applicability. Below is a concise comparison, focusing on advantages and disadvantages relative to XPS (based on 2025 standards, where XPS now includes microfocus and cluster-ion enhancements).

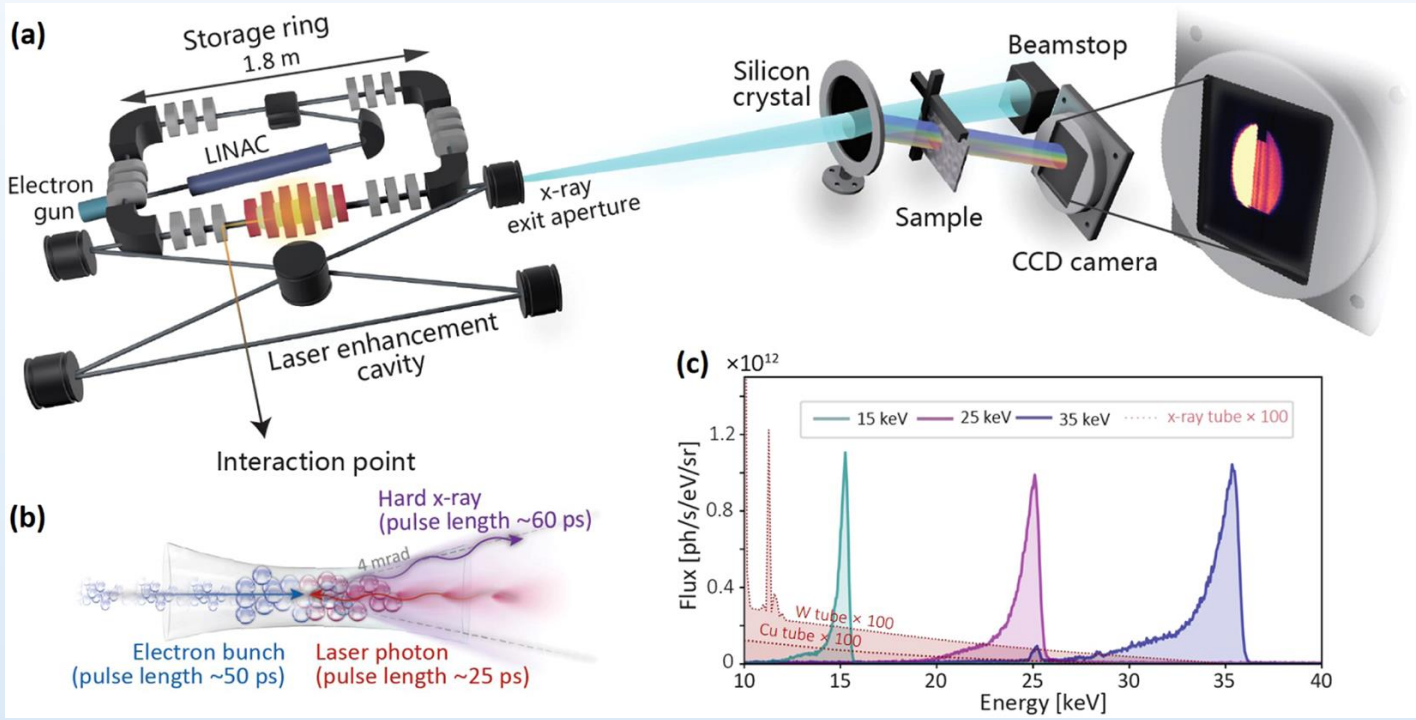
XPS remains superior for precise surface chemistry (e.g., bonding in thin films), but AES wins for high-res imaging, and EDS for rapid bulk screening. In 2025, hybrid systems (e.g., XPS+EDS) are common for comprehensive analysis.



TECHNIQUE	DESCRIPTION	ADVANTAGES OVER XPS	DISADVANTAGES VS. XPS
AUGER ELECTRON SPECTROSCOPY (AES)	Uses an electron beam to excite Auger electrons from the surface (~1–5 nm depth), providing elemental mapping.	- Higher spatial resolution (<10 nm) for micro-scale imaging. - Faster for spot analysis in scanning mode. - Better for insulators with less charging issues.	- Limited chemical-state info (no clear oxidation states). - More sample damage from electron beam. - Requires UHV
ENERGY-DISPERSIVE X-RAY SPECTROSCOPY (EDS/EDX)	Detects characteristic X-rays from electron-beam excitation, often integrated with SEM/TEM for bulk-to-surface elemental analysis (~1 µm–mm depth).	- Much faster scans (seconds vs. minutes for XPS). - Excellent spatial resolution (down to nm in SEM mode). - No UHV needed; works on large/bulk samples.	- No chemical-state sensitivity (can't distinguish bonds/oxidation). - Deeper penetration masks surface info. - Lower elemental detection limits (~0.1 wt% vs. XPS's 0.1 at%).



Auger Electron Spectroscopy (AES)



Energy-Dispersive X-ray Spectroscopy (EDS/EDX)

Available XPS Facilities in Iran

As of late 2025, Iran has a growing but limited number of X-ray Photoelectron Spectroscopy (XPS) facilities, primarily concentrated in major universities and select private labs. These centers support surface analysis for materials science, nanotechnology, and semiconductor research. Access is often through centralized laboratory services, with some offering paid external testing.





شرایط آنالیز XPS در آزمایشگاه تابان

زمان پاسخدهی آزمون XPS در آزمایشگاه بیم گسترتابان حدود یک تا دو ماه کاری پیش بینی می شود .

دستگاه مورد استفاده SPECS مدل UHV analysis system ساخت کشور آلمان می باشد .برای ارسال نمونه به موارد ذیل توجه گردد:

۱- نمونه لایه ای باید دارای ابعاد حداقل ۰.۵*۰.۵ سانتیمتر و حداکثر ۱.۵*۱.۵ سانتیمتر و ضخامت حداکثر ۵ میلیمتر باشند (پشت نمونه باید با علامت ضربدر مشخص شود)

۲- نمونه های پودری در آزمایشگاه بصورت قرص آماده سازی می شوند (پودر باید به اندازه ای باشد که یک سطح ۱.۵*۱.۵سانتیمتر را پوشش دهد)

۳- نمونه های مایع باید روی یک لام ۱*۱ سانتیمتر در چندین مرحله چکانده شود بصورتی که زیر لایه کاملا پوشانده شود (این عمل در آزمایشگاه انجام میشود در این حالت بگنواختی مایع مهم است)

آزمایشگاه بیم گستر تابان آماده ارائه خدمات آزمون XPS می باشد.





https://lims.labs.sharif.ir/landing/experiments-list?search=xps

https://www.kratos.com/product/axis-supra-xps-surface-analysis-instrument/

https://beamgostar.ir/xps-analysis/

Facility / Institution	Location	XPS Equipment / Model	Contact / Notes
Sharif University of Technology (Laboratory Services Center)	Tehran	BESTEC EA10 (monochromatic Al Kα source, hemispherical analyzer)	labs.sharif.ir Services resumed Aug 2024 after repairs; ISO 17025 accredited. External users welcome (fee-based).
University of Isfahan (Central Laboratory)	Isfahan	SPECS or similar UHV system (details limited)	Central Lab, University of Isfahan Active as of 2025; internal/external access via application.
BimGostar Taban Co. (Private Lab)	Tehran	SPECS UHV Analysis System (high-resolution HSA, multi-channel detector)	bimgostartaban.com Commercial services; outsourcing option for complex analyses.
Razi Metallurgical Research Center (Iranian Metallurgical Society)	Tehran	Integrated XPS with XRD/SEM (model not specified; likely older twin-anode)	razimetal.ir Government-affiliated; primarily internal use, but open to collaborators.
Institute for Research in Fundamental Sciences (IPM) (Nanophysics Lab)	Tehran	Custom synchrotron-linked XPS (tender-XPS capabilities via ILSF collaboration)	ipm.ac.ir Newer setup (2024–2025); focused on fundamental research, limited external access.

XPS Instrument Costs

Purchase Price

- **Entry-Level Systems**: Basic benchtop or twin-anode models (e.g., used/refurbished like older PHI systems) start at ~$50,000–$400,000 USD. These suit routine academic use but lack advanced features like imaging.
- **Mid-to-High-End Modern Systems**: Monochromatic Al Kα models with depth profiling and micro-XPS (e.g., Thermo K-Alpha, Kratos AXIS Supra, or Ulvac-PHI VersaProbe) range from $400,000–$1,500,000+ USD. Advanced options (e.g., with cluster-ion sources or ARXPS) push toward the upper end.
- **Factors Influencing Price**: Includes monochromator, detector type (MCP/PSD), software, and add-ons like vacuum transfer modules. Prices exclude installation (~10–20% extra) and are higher in sanctioned regions like Iran due to import challenges.

Maintenance and Operating Costs

- **Annual Maintenance Contracts**: 8–12% of purchase price (~$30,000–$150,000/year for high-end systems), covering calibration, parts (e.g., anodes, filaments), and on-site service. Essential for UHV stability and detector longevity.
- **Ongoing Operational Expenses**:
  - Consumables (power, cooling water, helium for cryopumps): $2,000–$10,000/year.
  - Total yearly cost: 10–15% of initial price for active labs, subsidized in universities (e.g., UCR model).
- **In Iran Context**: Local servicing limited; outsourcing to Europe/Asia adds 20–50% to costs, per Sharif University reports.

Low-Energy Electron Diffraction (LEED)

Introduction

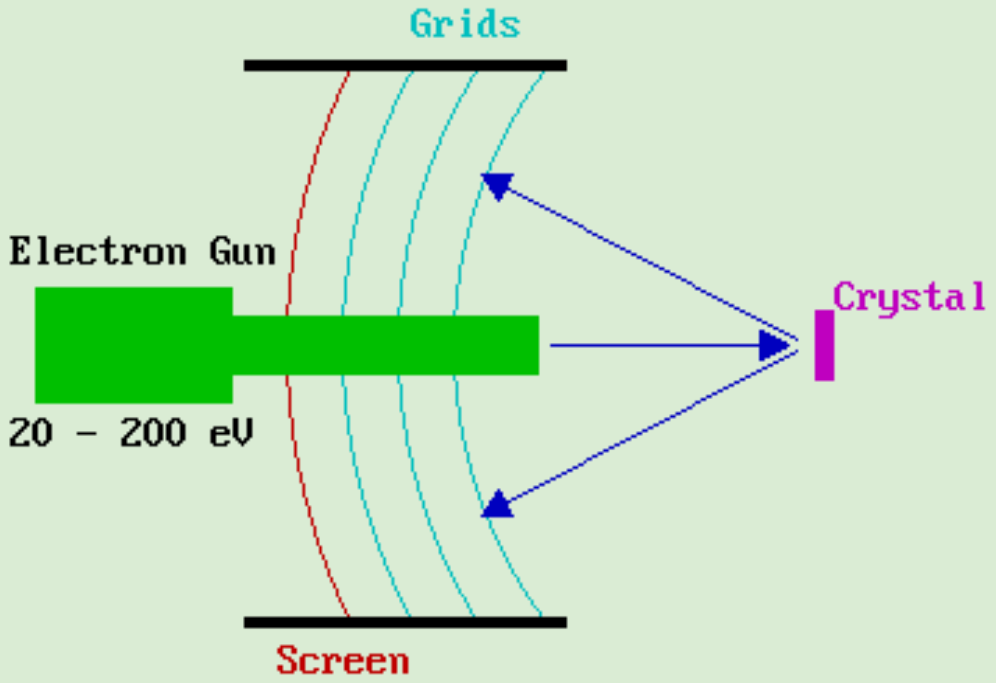
Low-Energy Electron Diffraction (LEED) is one of the most powerful techniques for directly determining the *atomic structure* of *single-crystal surfaces*.

A monochromatic beam of low-energy electrons (20–200 eV) is incident on the surface.

Due to their low energy, these electrons probe only the top few atomic layers, making LEED highly surface-sensitive.

Elastically backscattered electrons form a characteristic pattern of sharp spots on a fluorescent screen, reflecting the two-dimensional periodic arrangement of surface atoms.

Since the 1960s, LEED has been the gold standard in surface science for studying clean surfaces, adsorbates, and thin films, and remains essential in catalysis, 2D materials, and nanotechnology research.



History

In 1924 Louis de Broglie postulated that all forms of matter, such as electrons, have a wave-particle nature. Three years later, the American physicists Clinton J. Davisson and Lester H. Germer proved experimentally the wave nature of electrons at Bell Labs in New York.

Developed as a surface analysis method in the 1950s–1960s.

Introduction of ultra-high vacuum (UHV) systems made LEED practical for clean surfaces.

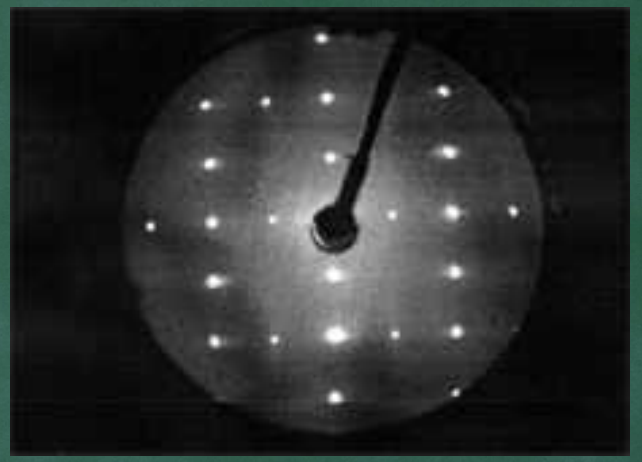
By the 1970s, LEED became a key technique for determining surface crystal structure.

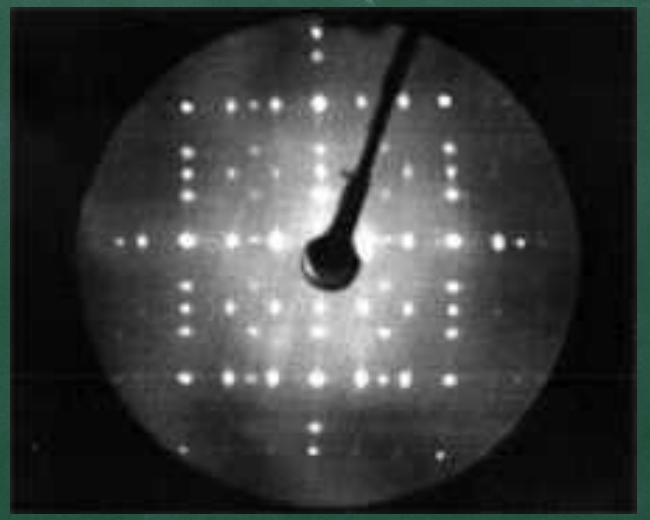
Still widely used as a standard tool in surface science



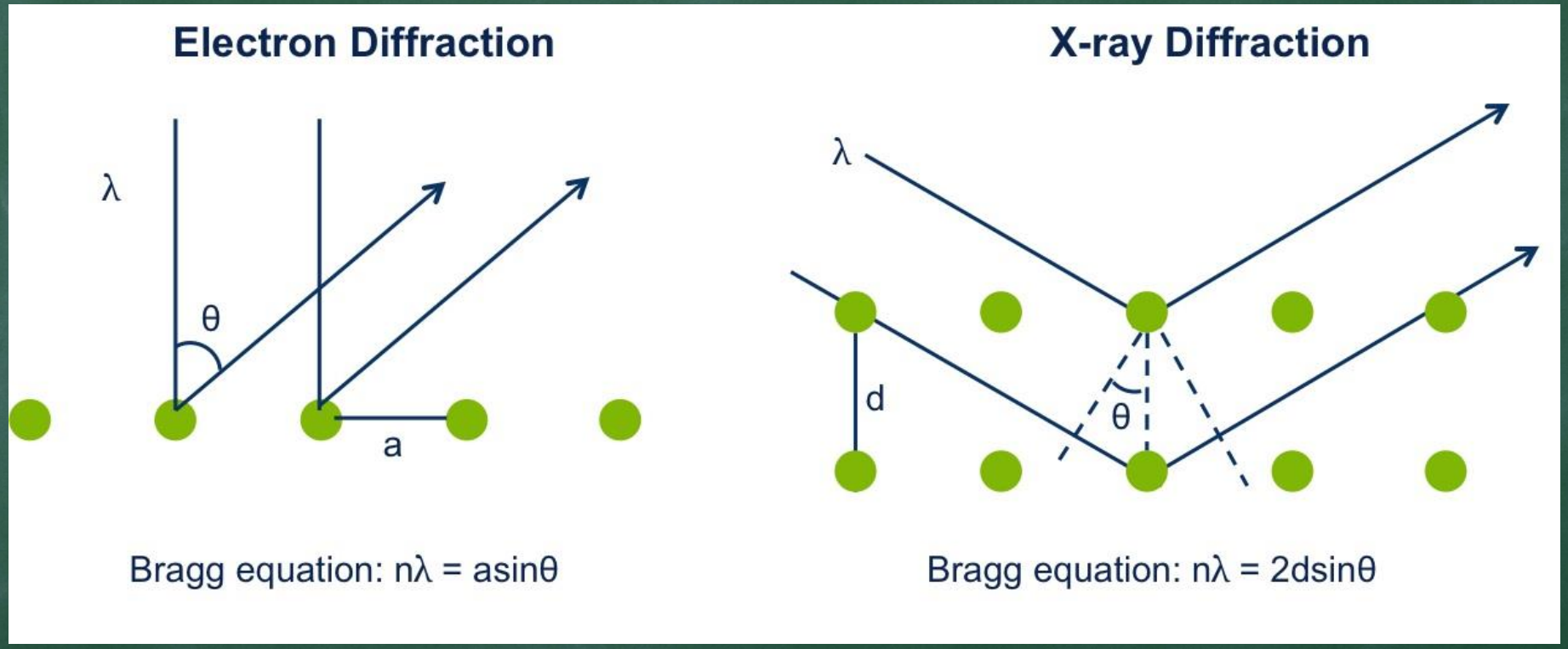
Principles and Diffraction Patterns

- 1. The Wave Nature (De Broglie Hypothesis):
  - Electrons behave as waves. $\lambda = h/(mv)$
  - The electron wavelength matches the inter-atomic spacing of crystals, making diffraction possible.
- 2. Why "Low Energy"? (Surface Sensitivity):
  - Energy range: 20 – 200 ev
  - Short Mean Free Path: Electrons penetrate only the skin of the crystal (5 – 10 A).
- 3. The Diffraction Condition (Bragg's Law):
  - Elastic Scattering: Only electrons that bounce back without losing energy contribute to the pattern.
  - Constructive Interference: Bright spots appear where scattered waves satisfy Bragg's Law.





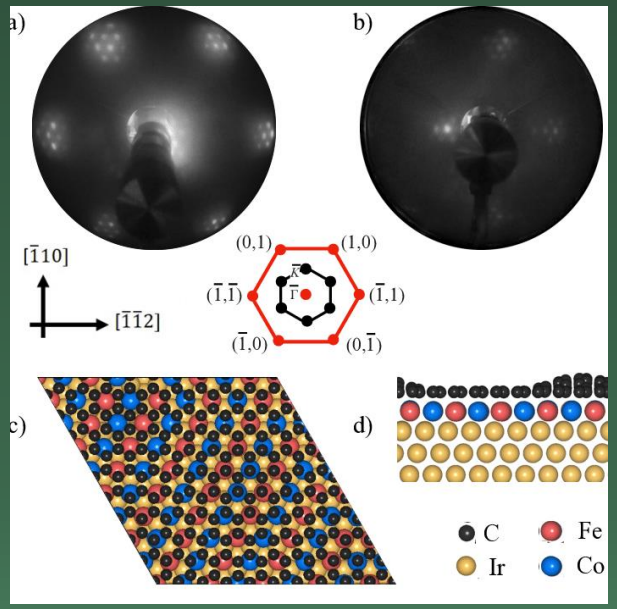
Like X-ray diffraction, electron diffraction also follows the Bragg's law

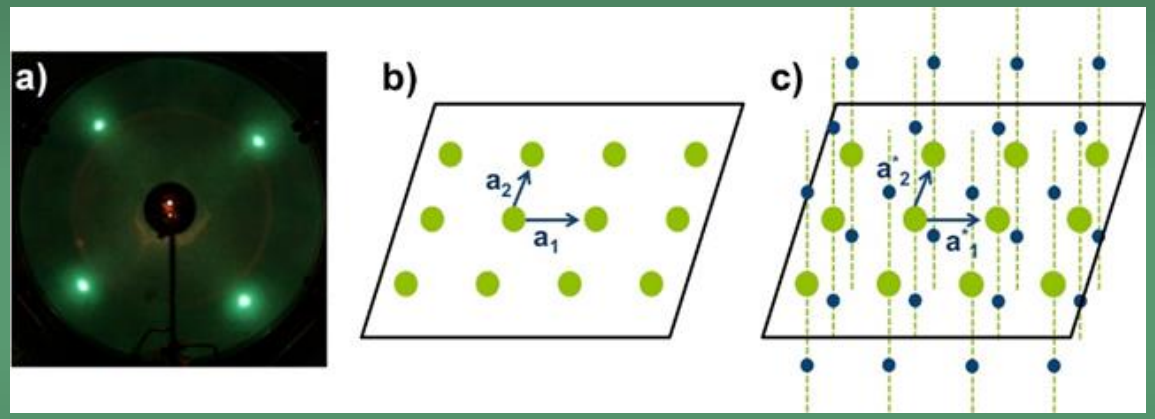


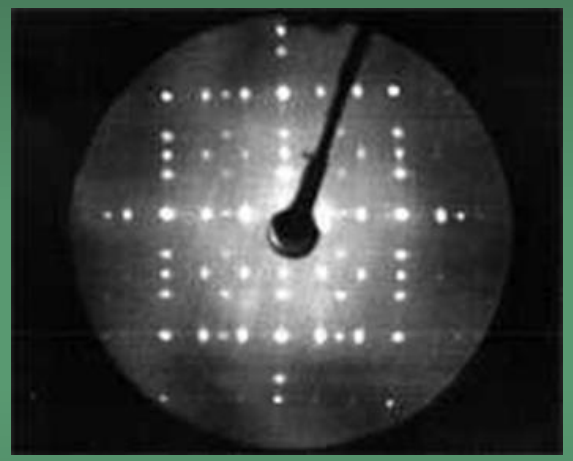
In LEED, the diffracted beams impact on a fluorescent screen and form a pattern of light spots, which is a to-scale version of the reciprocal lattice of the unit cell.

LEED patterns can give information about the size and shape of the real space unit cell, but nothing about the positions of the atoms. To gain this information about atomic positions, analysis of the spot intensities is required.

The separation of the points in the reciprocal lattice and the real interplanar distance are inversely proportional.







Experimental Equipment

The Requirement for Ultra-High Vacuum (UHV)

- **Pressure:** Experiments require extremely low pressure, typically $\leq 10^{-10}$ mbar (or 10^{-8} Pa).
- **Why UHV is Critical?**
  1. **Electron Gun Operation:** Necessary for the stability and lifetime of the electron source (filament).
  2. **Surface Cleanliness:** Prevents gas atoms (like O_2 or H_2) from immediately contaminating the sample surface.

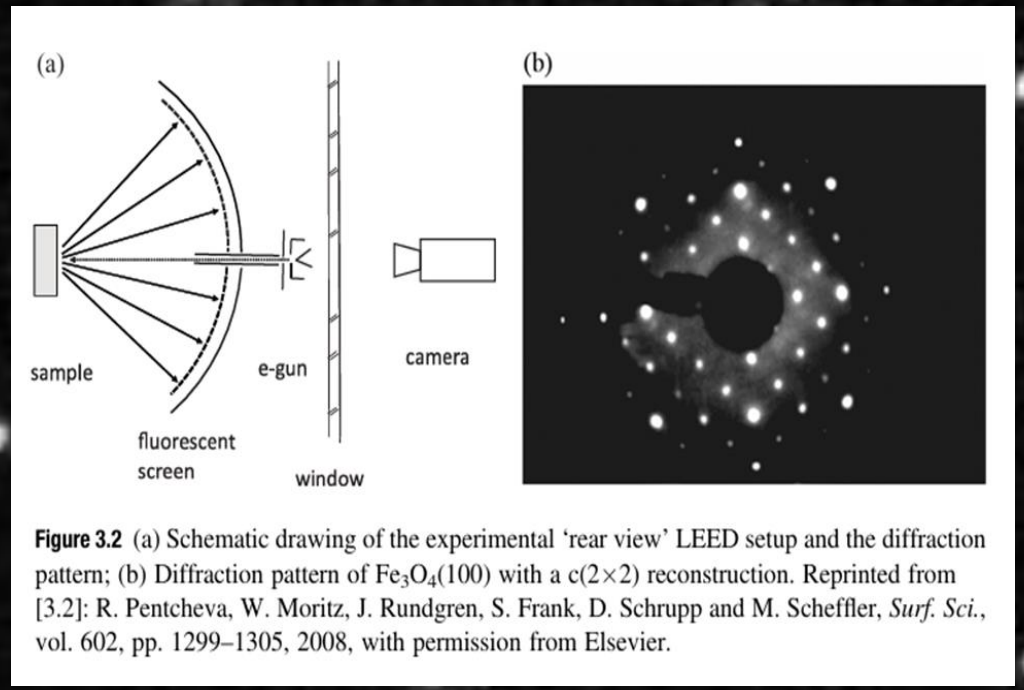


Figure 3.2 (a) Schematic drawing of the experimental 'rear view' LEED setup and the diffraction pattern; (b) Diffraction pattern of $Fe_3O_4(100)$ with a $c(2\times2)$ reconstruction. Reprinted from [3.2]: R. Pentcheva, W. Moritz, J. Rundgren, S. Frank, D. Schrupp and M. Scheffler, *Surf. Sci.*, vol. 602, pp. 1299–1305, 2008, with permission from Elsevier.

Equipment

- **Electron Gun:** Generates a highly monochromatic (single-energy) electron beam directed at the sample.
- **Sample Holder:** The sample is mounted on a holder that allows for precise positioning and manipulation.
- **LEED Optics:** The LEED optics focus and accelerate the electron beam and then direct the diffracted electrons towards a fluorescent screen.
- Fluorescent Screen: The fluorescent screen displays the diffraction pattern of the scattered electrons.
  The arrangement of the spots on the screen reveals the symmetry and periodicity of the crystal lattice

Mechanism

- **Electron Gun:** Generates a highly monochromatic (single-energy) electron beam directed at the sample.
- **Sample (Target):** Usually grounded (0 V). The electron beam is diffracted by the surface atoms.
- **System Layout (Reverse-View):** The detector (fluorescent screen) is located behind the electron gun. Diffracted electrons return to the detector in a backscattering geometry.
- **Screen Geometry:** The screen is typically spherical to ensure that the resulting diffraction pattern (the reciprocal lattice) is observed without distortion.

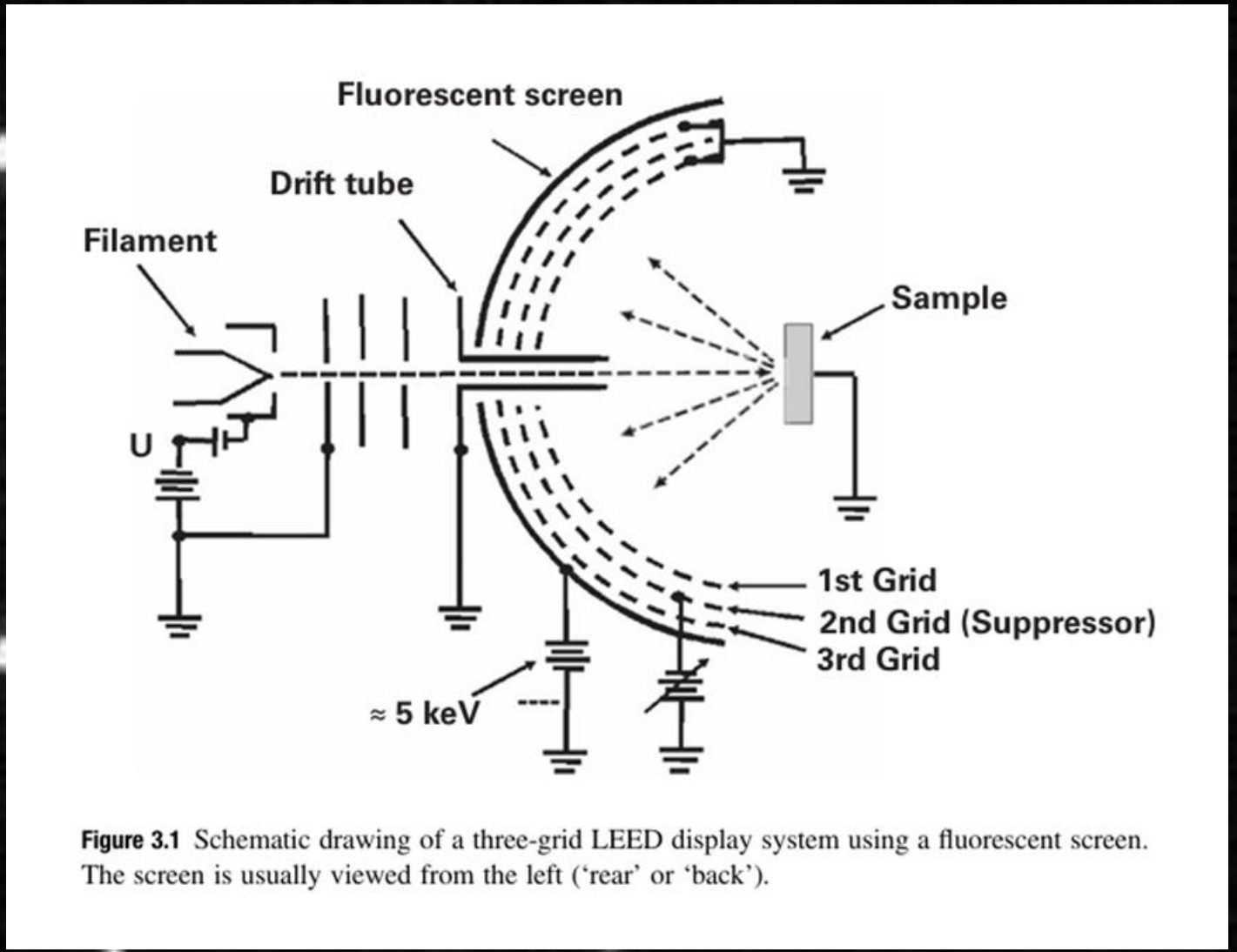


Figure 3.1 Schematic drawing of a three-grid LEED display system using a fluorescent screen. The screen is usually viewed from the left ('rear' or 'back').

Filtering Diffracted Electrons

The multi-grid system acts as an electrostatic high-pass filter to distinguish between valuable and irrelevant electrons.

the first grid are grounded, so the diffracted beams propagate straight through the space between the sample and the first grid.

The second and third grid blocks inelastically scattered electrons (those that lost energy, considered noise).

The filter voltage is slightly below the electron energy, typically a few eV, and adjustable.

There are three-grid and four-grid systems in use.

Fluorescent Screen Accelerates the filtered electrons to impact and cause fluorescence (light spots). High Positive Voltage (+4 to +7 kV)

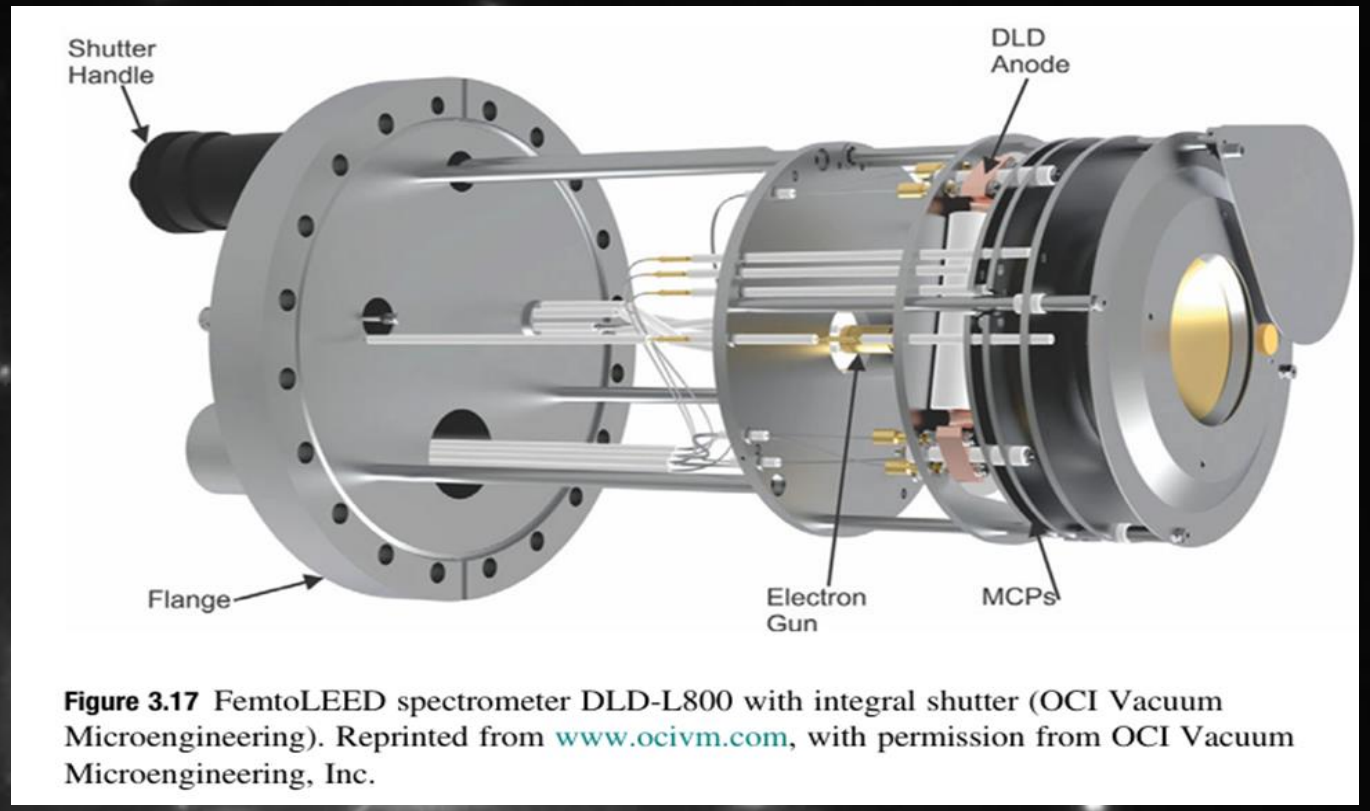


Figure 3.17 FemtoLEED spectrometer DLD-L800 with integral shutter (OCI Vacuum Microengineering). Reprinted from www.ocivm.com, with permission from OCI Vacuum Microengineering, Inc.

Diffraction Geometry

The reciprocal lattice of the 2-D periodic surface is a 2-D lattice of rods oriented perpendicular to the surface plane. The Ewald construction for elastic scattering shows that a reflection occurs whenever the Ewald sphere intersects a lattice rod.

The Ewald Sphere is a geometric tool used to determine which diffraction beams are generated.

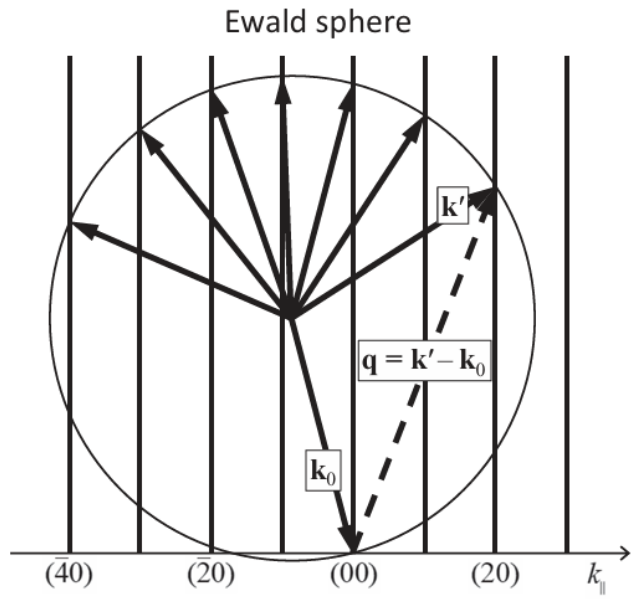


Figure 3.3 Diffraction geometry with Ewald construction of scattering directions of diffracted beams; the surface is horizontal in this representation, with $\mathbf{k}_0$ the incident beam momentum (in the plane of the figure) and $\mathbf{k}'$ one of several possible diffracted beam momenta (the Ewald sphere has radius $|\mathbf{k}_0|$); $\mathbf{q}$ gives the momentum transfer. The vertical lattice rods are given by the reciprocal lattice of the surface lattice.

Dynamic Effects:

- Changing Energy: Increasing voltage enlarges the sphere, causing spots to move toward the center.
- Changing Angle: Tilting the sample rotates the sphere relative to the rods, causing new spots to appear or disappear.

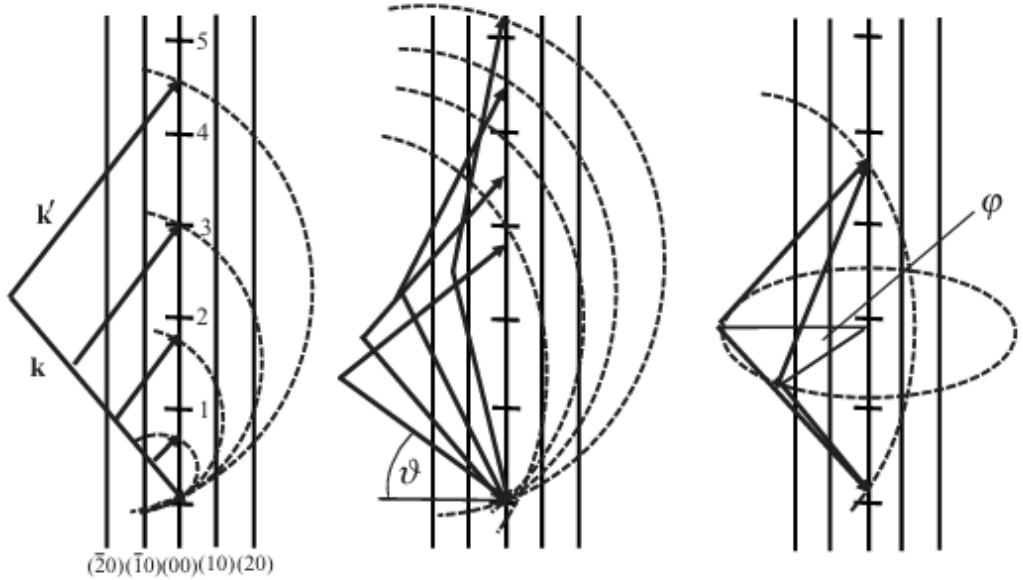


Figure 3.4 Measurement of I(V) curves (left panel), I(ϑ) curves (middle panel) and I(φ) curves (right panel). LEED intensities depend on the polar incidence angle ϑ and on the azimuthal angle φ. The same point in reciprocal space can be reached with different ϑ and change of energy and also with different azimuthal angle φ. In each case different intensities are measured, which is a multiple scattering effect. For kinematical diffraction, the usual case for X-ray diffraction, such dependence would not occur.

Think of the LEED pattern as a map of the surface atoms. The bright main spots represent the basic atomic structure. If we add a new layer of atoms in a larger pattern (a superlattice), new smaller spots appear between the main ones. In physics, we call these **fractional spots**. They are important because they act like a 'fingerprint,' telling us exactly how the atoms have rearranged themselves or how a new material, like Graphene, is growing on the surface.

The superlattice vectors (a', b') are √3 times longer than the substrate vectors (a, b).

Due to the inverse relationship, the new diffraction spots appear closer to the origin.

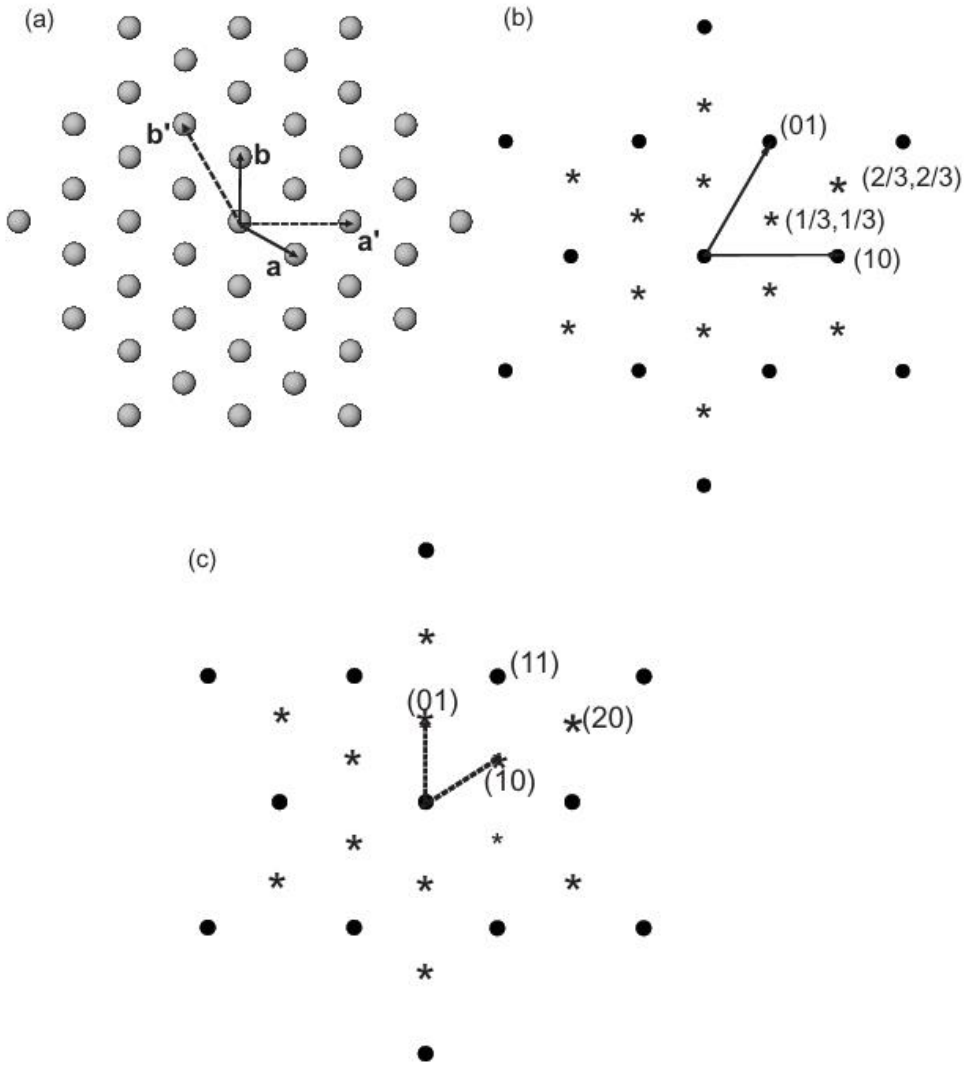


Figure 3.5 (a) Hexagonal lattice and unit cell vectors **a**, **b** for a (1×1) unit cell and **a′**, **b′** for a ($\sqrt{3}\times\sqrt{3}$)R30° superlattice; (b) reciprocal lattice with 'fractional' indices related to the (1×1) unit cell; and (c) with 'integer' indices related to the superlattice.

Pattern Recording and Data Analysis

- **Visual Output:** The bright spots on the screen correspond to the directions where constructive interference (diffraction) occurred.
- **Modern Detection:** The glowing pattern is recorded using high-sensitivity CCD or CMOS cameras.
- **Data Analysis (Intensity):**
  - **Qualitative:** Spot positions determine the size and shape of the unit cell.
  - **Quantitative (I-V Curves):** Spot intensities are measured as a function of the incident beam energy (I(V) curves). This data is necessary to determine the precise **atomic positions** on the surface.
- **Normalization:** Intensity measurements must be **normalized** to the primary beam current to account for any fluctuations in the electron source.

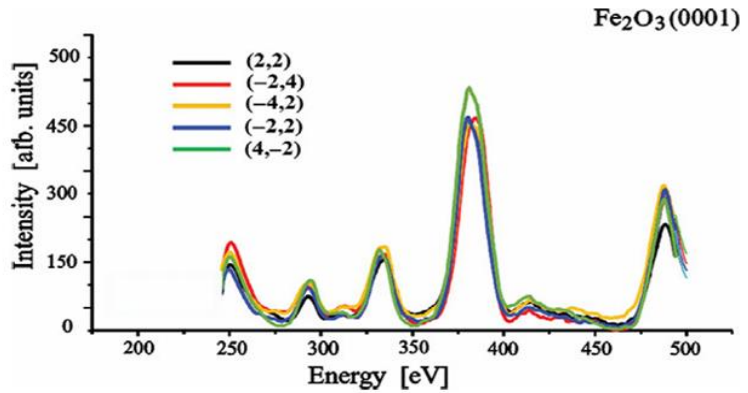


Figure 3.7 Intensities from five of six supposedly symmetrically equivalent beams near normal incidence from $Fe_2O_3(0001)$

Shielding and Beam Control

- **The Problem:** Low-energy electrons are easily deflected by external magnetic fields, even the Earth's field.
- **Solution (Shielding):** The entire experimental chamber must be enclosed in high-permeability material like **Mu-metal** to shield the electrons from external magnetic interference.
- **Alternative:** Using **Helmholtz coils** to generate compensating magnetic fields that neutralize the Earth’s field.

Maintaining a stable, undistorted electron path is crucial for achieving accurate crystallographic data.

Applications

Is the surface ordered? What is the structure of that order?

1. Surface Structure Determination:

- Confirms the crystalline order (sharp spots = ordered surface).
- Determines the symmetry and size of the surface unit cell.

2. Studying Surface Reconstruction:

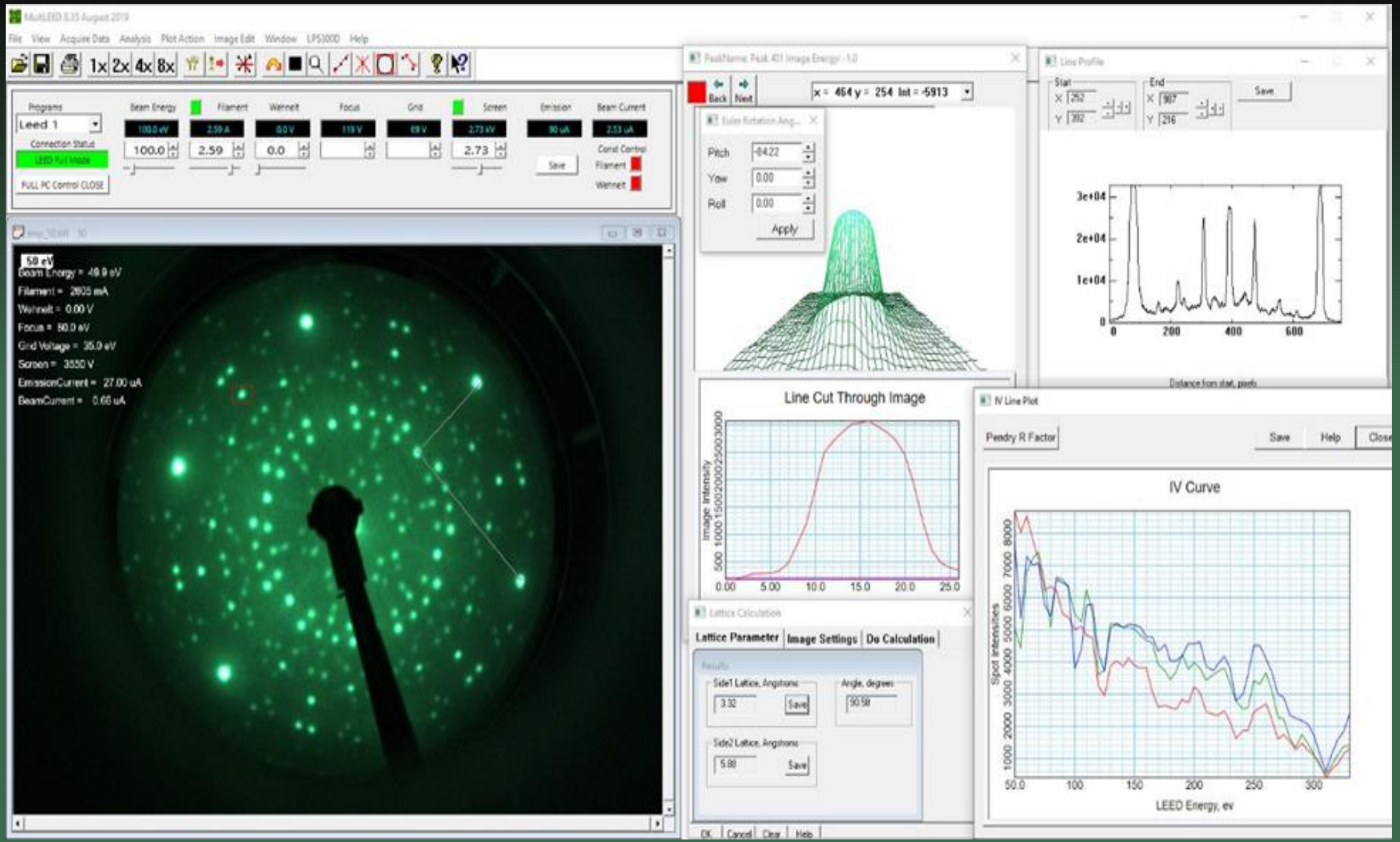
- Identifies the phenomenon where surface atoms rearrange to minimize energy (e.g., Si(111) – (7 × 7))
- Observed through extra diffraction spots in the pattern.

3. Adsorption and Overlayer Studies:

- Examines the structure of atoms or molecules (adsorbates) sitting on a substrate surface.
- Determines the adsorbate's unit cell and its location (registry) relative to the substrate.

4. Monitoring Thin Film Growth (Epitaxy):

- Used *in-situ* (during growth) to confirm that new layers are forming with high crystal quality and the desired structure . (MBE)


MuhLEED 3.15 August 2019
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LEED and Spot pattern

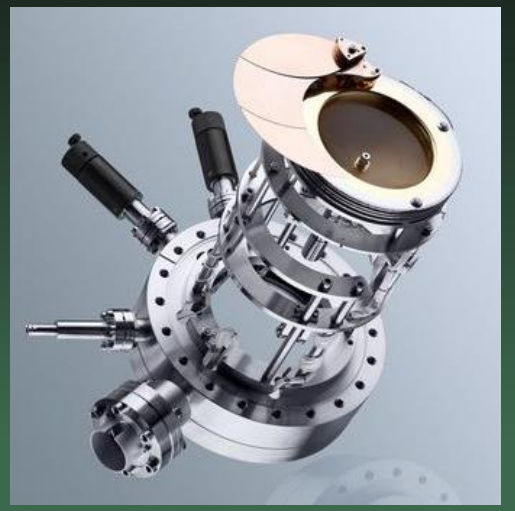
- In LEED we use less energy than TEM

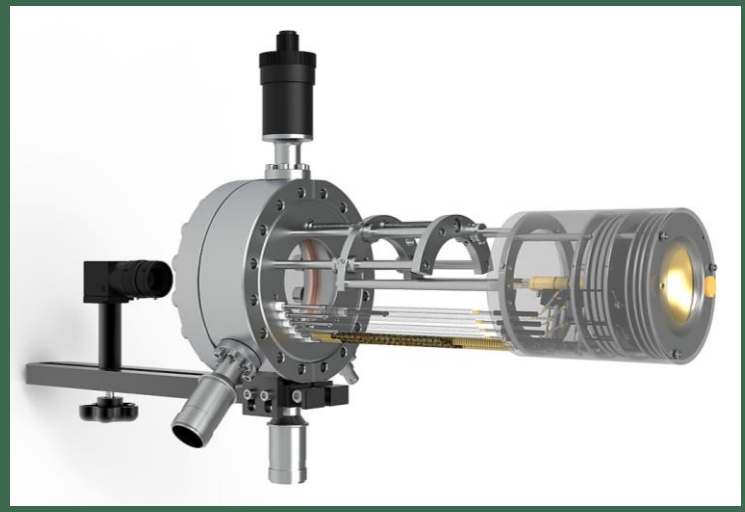
- In spot pattern we have diffraction only in planes which obey $hu + kv + lw = 0$, but in LEED we don't have zone axis rule.

- In TEM beam passes through the sample, but in LEED it doesn't.

Advantages & Disadvantages

Advantages	Disadvantages
High Surface Specificity: Only probes the top 1-5 atomic layers.	Requires Ultra-High Vacuum (UHV): Essential for maintaining a clean surface.
Simple Qualitative Check: Provides a fast, direct measure of surface order and symmetry.	Needs Single Crystal Samples: Only works on highly ordered surfaces.
Precise Quantitative Geometry: I(V) analysis determines interlayer spacing perpendicular to the surface.	Complex Quantitative Analysis: Requires demanding dynamical diffraction theory and computation.





Costs

1. Capital Cost (Purchase)

• Standalone LEED system: **USD 200,000 – 500,000**

• LEED integrated into a UHV surface analysis system (with XPS/AES, etc.): **USD 500,000 – 1,200,000+**

•Price depends on configuration, UHV chamber, electron gun, optics, control electronics, and service contract.

2. Operating Costs

• Hourly user fee (shared facility, if applicable): **USD 50 – 300+ per hour**

3. Maintenance & Consumables

•Annual maintenance/service: **USD 10,000 – 50,000**

4. Sample Preparation Costs

•UHV-compatible sample holders and cleaning tools: **USD 1,000 – 10,000**





Thank you

Mania Taheri
Sayena Soleimani
Diana Sohrabi